Must Any Consistent Physics Contain Structure Inaccessible to All Internal Observers?

A Gödel-Type Investigation of Whether Self-Consistent Physical Systems Necessarily Harbour Ontological Content Invisible from Within

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Abstract

We investigate whether every self-consistent physical system necessarily contains ontological content that is inaccessible to all observers situated within the system itself. Rather than adopting any single existing framework—be it Gödelian metamathematics, quantum-gravitational holography, autopoietic systems theory, or topos-theoretic physics—we develop an independent formal apparatus built from three primitive notions: *physical system*, *internal observer*, and *ontological content*. Consistency is defined as a conjunction of four conditions (logical, dynamical, informational, and self-referential), each of which is precisely characterised and shown to carry distinct formal consequences.

The mathematical core of the inaccessibility thesis rests on a structural fact: a proper part cannot biject onto the whole. Our contribution is to formalise this intuition rigorously, to explore its ramifications across multiple mathematical frameworks, to identify the precise boundary conditions under which it holds and fails, and—crucially—to distinguish the elementary cardinality obstruction from genuinely deeper phenomena that emerge at the self-referential and dynamical levels.

We pursue the investigation as a genuine mathematical expedition with no predetermined conclusion. First, we attempt to establish the inaccessibility thesis via five methods: a diagonal self-reference argument, an information-theoretic cardinality argument, a categorical fixed-point argument, a dynamical observational-equivalence argument, and a topological measure-theoretic argument. These methods share a common root in the non-injectivity of the representation map, but each reveals distinct structural consequences—from quantitative

bounds on the accessibility ratio, through temporal persistence of equivalence classes, to generic inaccessibility in the measure-theoretic sense. We are explicit about which results are elementary reformulations of cardinality constraints and which constitute genuinely independent insights.

Second, we subject this thesis to sustained attack by attempting to prove its negation through four strategies: the trivial-system construction, a holographic boundary-access argument, a collective-observer covering argument, and a systematic relaxation of each consistency condition. Third, we examine the possibility that the question is itself undecidable from within, employing self-referential and model-theoretic independence arguments.

To ground the abstract framework in concrete settings, we instantiate the formalism in three toy models—a cellular automaton with an embedded observer agent, a coupled oscillator system, and a miniature formal system—and additionally develop a quantum-mechanical instantiation using projection-valued measures as the physically natural property class. We investigate whether the inaccessibility thesis survives restriction from arbitrary subsets to physically motivated property classes (Borel-measurable sets, projection operators, macroscopic observables) and find that it does, though the quantitative gap depends on the property class.

Every definition, proposition, and theorem is preceded and followed by thorough explanatory discussion to ensure accessibility to readers whose primary training lies outside dynamical systems and mathematical logic. All computational code is provided in the appendices.

The paper reports what the mathematics yields, with minimal philosophical extrapolation beyond the formal results.

Contents

1 Introduction

1.1 The Central Question

Consider a physical system—any physical system—that is self-consistent in the sense that its description harbours no internal contradictions. Now suppose an observer exists *within* that system: not looking at it from some external vantage point, but constituted by the system's own degrees of freedom, interacting with the rest of the system through the system's own laws.

The question this paper investigates is:

Must such a system contain facts about itself that no internal observer, however capable, can ever access?

This is not a question about practical limitations—about finite memory, thermal noise, or the particular structure of human cognition. It is a question about *structural necessity*: whether the very conditions that make a physical system self-consistent force the existence of ontological content that is invisible from within.

The question has obvious ancestors. Gödel's incompleteness theorems [1] established that any sufficiently powerful consistent formal system contains true statements it cannot prove. Turing's halting theorem [2] showed that no algorithm can decide an arbitrary algorithm's behaviour. In physics, the development of black hole complementarity [3], the holographic principle [4, 5], and recent no-go results concerning observer-dependent facts in quantum mechanics [6, 7] all point toward fundamental limits on what observers can know from within their physical setting. The programme of quantum Darwinism [8, 9] and decoherence theory provides perhaps the most developed existing framework for understanding how internal observers acquire information about their environment and which properties become "objective" (accessible to multiple observers). Wolpert's results on physical limits of inference [10] similarly address the question from an information-theoretic perspective. Most directly relevant is Breuer's proof [11] that no quantum subsystem can perform a complete state self-measurement—a result that establishes, within the specific framework of quantum mechanics, the impossibility that this paper investigates in a theory-independent setting. Svozil [12] has explored related themes of intrinsic indeterminacy and undecidability in physical systems.

Yet none of these results, individually or collectively, settles the question as posed. Gödel's theorems apply to formal systems of a particular kind, not to arbitrary physical systems. The physical results depend on specific physical theories (quantum mechanics, general relativity) and specific notions of observation. Quantum Darwinism identifies *which* properties become classically accessible but does not address the general structural question of whether inaccessible content must exist. What is missing is a framework that is simultaneously:

1. **General**—applicable to any self-consistent physical system, not merely those described by current theories;

2. **Precise**—with rigorous definitions and complete proofs, not analogies or heuristic arguments;

3. **Self-critical**—genuinely testing the claim against its own negation, rather than constructing a one-sided case.

This paper attempts to supply such a framework. We should state at the outset what the mathematical core of the investigation reveals: the inaccessibility thesis rests, at its root, on the structural fact that a proper part cannot biject onto the whole. Our contribution is not the discovery of this fact but its rigorous formalisation in a theory-independent setting, the exploration of its consequences across multiple mathematical frameworks, the precise identification of boundary conditions (the trichotomy theorem), and the demonstration that the result persists for physically motivated property classes. We are explicit throughout about which results are elementary reformulations of cardinality constraints and which constitute genuinely independent insights.

1.2 What This Paper Does and Does Not Do

We develop an independent formal apparatus built from three primitive notions—*physical system*, *internal observer*, and *ontological content*—and a four-part definition of *consistency* (logical, dynamical, informational, and self-referential). From these definitions alone, without postulating axioms, we derive all subsequent results.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. We pursue three lines of inquiry in sequence:

1. **The Inaccessibility Thesis.** We attempt to prove, by five independent methods, that any consistent physical system containing an internal observer necessarily harbours ontological content inaccessible to that observer.

2. **The Accessibility Antithesis.** We then attempt, with equal seriousness, to prove the negation: that there exist consistent physical systems in which internal observers can access the totality of ontological content.

3. **The Undecidability Possibility.** Finally, we investigate whether the question might be undecidable—neither provable nor refutable within the framework—which would itself constitute a result of considerable depth, given the paper's subject matter.

The paper reports what the mathematics yields, with minimal extrapolation beyond the formal results. Where observations venture into interpretive territory, they are clearly flagged as such. Readers from all disciplines are welcome to draw their own conclusions; the paper's task is to make the formal landscape clear enough to support such reflection.

1.3 Relation to Existing Work

Four intellectual traditions bear on the present question. We engage each seriously but develop a framework independent of all four.

Gödelian metamathematics and its extensions. The incompleteness theorems [1] and their extensions by Rosser [13], Löb [14], and others establish that sufficiently expressive consistent formal systems are necessarily incomplete. Lucas [15] and Penrose [16, 17] have drawn conclusions about the nature of mind from these results. Our work differs in that we do not assume the physical system is a formal system of arithmetic, nor do we import conclusions about consciousness. Our diagonal argument (Section 4) is a construction within our own framework, not an application of Gödel's theorems.

Observer-dependent limits in physics. Bohr's complementarity principle, the Heisenberg uncertainty relations, black hole complementarity [3], the holographic bound [4, 5], and recent no-go theorems on observer-independent facts [6, 7] all identify specific physical mechanisms that limit internal observation. Our framework is not tied to any specific physical theory: the dynamical and topological arguments (Sections 7 and 8) recover *structural* analogues of these results from the definitions alone, without invoking quantum mechanics or general relativity.

Self-reference in cybernetics and systems theory. The traditions of von Foerster [18], Maturana and Varela [19], Luhmann [20], and more recently Friston [21] address how self-referential systems create and maintain internal boundaries that limit self-observation. Our treatment of self-referential consistency (Section 3) formalises a version of these ideas without adopting the biological or sociological vocabulary.

Topos-theoretic and categorical approaches. Döring and Isham [22, 23], and subsequent work in categorical quantum mechanics [24, 25], reformulate physical theories within topoi where truth values are contextual. Our fixed-point argument (Section 6) employs category-theoretic tools—specifically Lawvere's fixed-point theorem [26]—but the framework itself is not topos-theoretic.

The present work charts a fifth path. It shares concerns with all four traditions but derives its results from its own definitions.

1.4 Guide for the Reader

The paper is constructed so that each section builds upon the previous ones, with definitions introduced precisely when they are needed by the next proof. The structure is as follows.

Section 2 introduces the three primitive notions: physical system, internal observer, and ontological content. Section 3 develops the four-part definition of consistency. The inaccessibility thesis is then pursued through five proof methods in Sections 4 to 8. The accessibility antithesis is examined in Sections 9 to 12. The undecidability question is treated in Section 13. Section 14 synthesises the results, and Section 15 offers concluding remarks. Computational code for all toy models and figures appears in Appendix A. A placeholder for peer review and revision history is provided in Appendix B.

Each major step is accompanied by a boxed summary explaining its strategic purpose within the overall argument, and by a self-critique subsection that identifies potential weaknesses and either resolves them or flags them as open problems. Theorems are proved in full; where multiple proof methods are available, we present more than one.

Strategic Purpose

The introduction establishes three things. First, it poses the question in terms accessible to any scientifically literate reader. Second, it positions the paper relative to existing traditions without depending on any of them. Third, it sets the methodological ground rules: complete proofs, genuine self-critique, and no predetermined conclusion. This framing is necessary because the question sits at a junction of disciplines that rarely share a common formal language, and the paper must earn the trust of readers from each.

1.5 Self-Critique and Resolution

Critique 1: Overambition. The paper claims to investigate *all* consistent physical systems with a single framework. This is an extraordinarily broad claim. Can any finite formal apparatus truly capture such generality?

Resolution. The generality is achieved by minimality: the framework assumes very little about what a physical system is (a state space with dynamics) or what an observer is (a distinguished subsystem with a representation map). The less one assumes, the more broadly the results apply—but the less specific the conclusions. We are explicit throughout about what holds under the minimal definitions and what requires additional structure.

Critique 2: The "fifth path" framing. Claiming independence from four major traditions risks appearing either naïve or arrogant. The traditions exist for good reasons, and the paper's results must ultimately be relatable to them.

Resolution. Independence does not mean ignorance. Each proof method has a clear intellectual ancestor in one of the four traditions, and we identify these connections explicitly. The claim is that the formal definitions are self-standing—not that the ideas emerged in a vacuum.

Critique 3: No predetermined conclusion. If the paper genuinely does not know where it is going, how can it be structured coherently?

Resolution. The structure is that of a mathematical investigation, not a rhetorical argument. The thesis and antithesis are pursued with equal rigour, and the synthesis reports the outcome. This is standard in exploratory mathematics (compare, for example, the investigation of the Continuum Hypothesis before Cohen's forcing results). The paper's coherence comes from the consistency of its definitions and the completeness of its proof attempts, not from the direction of its conclusions.

2 Primitive Notions

This section introduces the three foundational concepts upon which the entire paper rests: *physical system*, *internal observer*, and *ontological content*. Each is defined with the minimum structure necessary for the subsequent proofs, and each definition is accompanied by discussion of what it captures and what it deliberately leaves open.

We do not postulate axioms. Instead, we define objects and derive their properties. If a theorem requires an additional condition, that condition is stated as a hypothesis of the theorem, not as a standing assumption.

2.1 Physical Systems

We begin with the most fundamental notion: a system that can be in various states and that evolves according to some rule.

Definition 2.1 (Physical System)**.** A *physical system* is a triple $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ where:

1. Σ is a non-empty set, called the *state space*;
2. t is a non-empty totally ordered commutative semigroup (with operation $+$ and order $\leq$), called the *time structure*;
3. $\mathcal{D} : \Sigma \times t \to \Sigma$ is a map, called the *dynamics*, satisfying the semigroup property: $\mathcal{D}(\mathcal{D}(\sigma, \tau_1), \tau_2) = \mathcal{D}(\sigma, \tau_1 + \tau_2)$ for all $\sigma \in \Sigma$ and $\tau_1, \tau_2 \in t$.

To appreciate this definition: the state space Σ gathers all the configurations the system could possibly occupy. It might be a finite set of configurations (as in a cellular automaton), a smooth manifold (as in Hamiltonian mechanics), a Hilbert space (as in quantum mechanics), or any other mathematical structure. The definition does not presuppose a particular type of physics.

The dynamics $\mathcal{D}$ encodes how the system changes. Given a state and an elapsed time, $\mathcal{D}$ produces the resulting state. The semigroup property ensures that evolution is self-consistent: evolving for time τ_1 and then for time τ_2 yields the same result as evolving for time $\tau_1 + \tau_2$. This property is what prevents the dynamics from contradicting itself across different time intervals.

The time structure t is required to be an ordered commutative semigroup—it must support an addition operation (to compose durations) and a total order (to sequence events). Standard examples include $\mathbb{N}$ (discrete time), $\mathbb{Z}_{\geq 0}$ (non-negative integer time), and $\mathbb{R}_{\geq 0}$ (continuous time). The algebraic requirement is not redundant: a mere total order would not guarantee that the semigroup property in condition (3) is well-formed, since $\tau_1 + \tau_2$ must be defined.

Remark 2.2 (Limitation: Total Ordering of Time). The requirement that t be totally ordered excludes *partially ordered* time structures, which arise naturally in relativistic physics where spacelike-separated events have no canonical temporal ordering. In a relativistic setting, the appropriate generalisation would replace the total order with a causal partial order (as in the causal set programme or the framework of globally hyperbolic spacetimes). The present framework therefore applies, strictly speaking, to systems with a global time parameter—which includes non-relativistic quantum mechanics, classical mechanics, and discrete-time systems—but not directly to generally covariant field theories. Extending the definitions to partially ordered time is a natural direction for future work and would require replacing the semigroup property with a more general composition law indexed by causal paths.

Definition 2.3 (Subsystem)**.** Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$, a *subsystem* is a triple $\mathcal{S}' = (\Sigma', \mathcal{D}', t)$ together with a pair of maps:

1. An *embedding* $\iota : \Sigma' \to \Sigma$ that is injective;
2. A *projection* $\pi : \Sigma \to \Sigma'$;

satisfying $\pi \circ \iota = \mathrm{id}_{\Sigma'}$ and the *compatibility condition*: for every $\sigma \in \Sigma$ and every $\tau \in t$,

$$\pi(\mathcal{D}(\sigma, \tau)) = \mathcal{D}'(\pi(\sigma), \tau).$$

A subsystem is a portion of the full system that can be meaningfully isolated. The embedding ι tells us how subsystem states sit inside the full state space. The projection π tells us how to extract the subsystem's state from the full system's state. The condition $\pi \circ \iota = \mathrm{id}_{\Sigma'}$ ensures that embedding a subsystem state and then projecting it back recovers the original subsystem state. The compatibility condition ensures that the subsystem's own dynamics agrees with the restriction of the full dynamics to the subsystem's degrees of freedom.

Notice what this definition does *not* require: it does not demand that the subsystem is dynamically independent or isolated from the rest of the system. The projection π may lose information—the full state may contain features not captured by the subsystem's state—and this is precisely the structure that will become important later.

Strategic Purpose

Defining physical systems as abstract triples $(\Sigma, \mathcal{D}, t)$ with a minimal semigroup property ensures that our results are not artifacts of a particular physical theory. Any system described by Newtonian mechanics, quantum mechanics, thermodynamics, cellular automata, or even exotic hypothetical dynamics is captured as a special case. The subsystem definition with its embedding–projection pair creates the mathematical language for asking: what does the subsystem "see" of the whole?

2.2 Internal Observers

An observer that exists *within* a system is a subsystem with additional structure: it has the capacity to form representations of the system's states.

Definition 2.4 (Internal Observer)**.** Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$, an *internal observer* is a quadruple $\mathcal{O} = (\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t, \mathcal{R})$ where:

1. $(\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t)$ is a subsystem of $\mathcal{S}$ with embedding $\iota_{\mathcal{O}}$ and projection $\pi_{\mathcal{O}}$;
2. $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ is a map, called the *representation map*, satisfying:
   (a) *Physicality*: There exists a fixed $\tau_0 \in t$ (the *measurement duration*) such that for every $\sigma \in \Sigma$, $\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_0)) = \mathcal{R}(\sigma)$. That is, the observer's representation of the system state is the result of letting the system evolve for a definite duration and reading off the observer's degrees of freedom. The duration τ_0 is uniform across all states.
   (b) *Internality*: $\mathcal{R}$ factors through $\pi_{\mathcal{O}}$. That is, $\mathcal{R} = f \circ \pi_{\mathcal{O}}$ for some map $f : \Sigma_{\mathcal{O}} \to \Sigma_{\mathcal{O}}$. The observer can only use its own degrees of freedom to build representations.

The physicality condition deserves careful attention. The requirement that τ_0 be *uniform*—the same for all σ—is essential. Without it, the observer could in principle select a different measurement duration for each state, which would presuppose knowledge of the state prior to measurement. The uniform-τ_0 version captures the physical reality that a measurement protocol is fixed before the system's state is known.

The *internality* condition says that the observer's representations are constructed entirely from its own degrees of freedom. The observer cannot access information that does not pass through its own state space. Together, these two conditions impose a fundamental constraint: the map $\mathcal{R}$ cannot be arbitrary. It must be realisable by the system's own dynamics at a definite time, and it must factor through the observer's own (typically smaller) state space.

Definition 2.5 (Observational Equivalence)**.** Given an internal observer $\mathcal{O}$ with representation map $\mathcal{R}$, two states $\sigma_1, \sigma_2 \in \Sigma$ are *observationally equivalent with respect to* $\mathcal{O}$, written $\sigma_1 \sim_{\mathcal{O}} \sigma_2$, if and only if $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$.

Observational equivalence is the relation that identifies states the observer cannot distinguish. Two states are observationally equivalent if the observer represents them identically. The equivalence classes under $\sim_{\mathcal{O}}$ partition the state space into groups of states that, from the observer's perspective, are indistinguishable.

This is a straightforward consequence of the definition: since $\mathcal{R}$ maps the full state space to the observer's (typically smaller) state space, it cannot in general be injective. Distinct system states may map to the same observer state, and such states are precisely those the observer cannot tell apart.

Definition 2.6 (Maximally Capable Observer)**.** An internal observer $\mathcal{O}$ is *maximally capable* if, among all internal observers of $\mathcal{S}$, its representation map $\mathcal{R}_{\mathcal{O}}$ has the finest equivalence relation. Formally, $\mathcal{O}$ is maximally capable if for every internal observer $\mathcal{O}'$ of $\mathcal{S}$, $\sigma_1 \sim_{\mathcal{O}} \sigma_2$ implies $\sigma_1 \sim_{\mathcal{O}'} \sigma_2$.

A maximally capable observer is one that can make the finest distinctions among system states that any internal observer can make. If even this observer cannot distinguish two states, then no internal observer can. This notion allows us to ask the strongest version of our question: does the maximally capable observer still have blind spots?

Strategic Purpose

The internal observer definition is the pivot of the entire paper. By requiring that the representation map be *physical* (arising from dynamics) and *internal* (factoring through the observer's own state space), we ensure that any limitations we prove are not due to artificial restrictions on the observer's capabilities, but follow from the structural fact of being embedded within the system one observes. The notion of observational equivalence provides the precise mathematical handle on "what the observer cannot see."

2.3 Ontological Content

We need to make precise what "all the facts about the system" means, so that we can ask whether an observer can access all of them.

Definition 2.7 (Property)**.** A *property* of a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is a subset $P \subseteq \Sigma$. A state σ *has property* P if $\sigma \in P$.

This is the most general possible notion of a property: any collection of states constitutes a property, namely the property of being in one of those states. The property "having energy greater than E_0" corresponds to the set of all states whose energy exceeds E_0. The property "being at equilibrium" corresponds to the set of equilibrium states. Even structural, relational, or higher-order features of the system can be encoded as properties in this sense, since any criterion that either holds or fails for a given state defines a subset of Σ.

Remark 2.8 (Properties and Propositions). Throughout this paper, "property" (a subset $P \subseteq \Sigma$) and "fact" or "truth about the system" (a proposition that the system is in a state belonging to P) are used in close correspondence but are distinct. A property is a *type-level* object: a collection of states. A fact is a *token-level* assertion: that a *specific* state σ belongs to P. When we say "the observer cannot access property P," we mean that the observer cannot determine whether the system's current state is in P—that is, the observer cannot evaluate the proposition "$\sigma \in P$" for an unknown σ. This distinction matters in philosophy of physics, where "properties of a system" and "facts about a system" may carry different ontological commitments. Our framework is neutral on this point: the formal results hold regardless of one's metaphysical interpretation of subsets.

Definition 2.9 (Ontological Content)**.** The *ontological content* of a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is the power set $\mathfrak{O}(\mathcal{S}) = \mathcal{P}(\Sigma)$: the collection of all properties of $\mathcal{S}$.

The ontological content is the totality of facts that could, in principle, be true of the system. It is maximal by construction: every possible property is included.

Definition 2.10 (Accessible Content)**.** Given an internal observer $\mathcal{O}$ with representation map $\mathcal{R}$, the *accessible content* is the set of properties that $\mathcal{O}$ can distinguish:

$$\mathfrak{A}(\mathcal{O}) = \{P \subseteq \Sigma \mid P \text{ is a union of equivalence classes of } \sim_{\mathcal{O}}\}.$$

Equivalently, $P \in \mathfrak{A}(\mathcal{O})$ if and only if $P = \mathcal{R}^{-1}(\mathcal{R}(P))$.

A property is accessible to the observer if and only if the observer can determine whether the system possesses that property. This happens precisely when the property respects observational equivalence: whenever a state has the property, all states observationally equivalent to it also have the property. If a property fails this condition—if it holds for some state σ_1 but not for some σ_2 with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$—then the observer cannot determine whether the system has this property, because it cannot distinguish σ_1 from σ_2.

The accessible content $\mathfrak{A}(\mathcal{O})$ is a sub-Boolean-algebra of $\mathcal{P}(\Sigma)$: it is closed under union, intersection, and complement, and contains $\varnothing$ and Σ. This is immediate from the fact that unions of equivalence classes are closed under these operations.

Definition 2.11 (Inaccessible Content)**.** The *inaccessible content with respect to* $\mathcal{O}$ is the complement

$$\mathfrak{O}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O}) = \{P \subseteq \Sigma \mid P \notin \mathfrak{A}(\mathcal{O})\}.$$

The *universally inaccessible content* is

$$\mathfrak{I}(\mathcal{S}) = \mathfrak{O}(\mathcal{S}) \setminus \bigcup_{\mathcal{O}} \mathfrak{A}(\mathcal{O}),$$

where the union ranges over all internal observers of $\mathcal{S}$.

The universally inaccessible content consists of properties that *no* internal observer of the system can determine. The central question of this paper can now be stated with full precision:

Central Question (Formal Version).

For which physical systems $\mathcal{S}$ satisfying the consistency conditions of Section 3, is $\mathfrak{I}(\mathcal{S}) \neq \varnothing$? Must it always be non-empty?

Strategic Purpose

Ontological content is defined as the full power set $\mathcal{P}(\Sigma)$ because this represents the maximal collection of distinctions that could be drawn within the system. Accessible content is the sub-collection an observer can detect, determined entirely by the observer's representation map. The gap between the two—if it exists—is the inaccessible content. This setup transforms a philosophical question ("are there things we cannot know from within?") into a precise mathematical question ("is $\mathfrak{I}(\mathcal{S})$ non-empty?"), which can be attacked with formal methods.

2.4 Self-Critique and Resolution

Critique 1: The power set is too large. Defining ontological content as $\mathcal{P}(\Sigma)$ includes pathological, non-measurable, and physically meaningless subsets of Σ. A more reasonable definition might restrict to measurable sets, Borel sets, or clopen sets, depending on the topology of Σ.

Resolution. This is deliberate. By including all subsets, we make the inaccessibility thesis *easier* to prove and the accessibility antithesis *harder* to prove. If inaccessibility fails even with the maximal definition of ontological content, it certainly fails for any restricted notion. Conversely, if accessibility holds for the maximal definition, it holds a fortiori for any restriction. In Section 8, we revisit this point with a more refined measure-theoretic formulation and show that the results survive restriction to physically natural classes of properties.

Critique 2: The representation map is deterministic. The definition requires $\mathcal{R}$ to be a function, producing a single observer state for each system state. Quantum mechanics suggests that measurement outcomes may be inherently stochastic.

Resolution. A stochastic representation can be modelled within this framework by enlarging the state space to include the random seed (or the full quantum state including environmental degrees of freedom). After this enlargement, the representation map is again deterministic on the enlarged state space. Alternatively, one can define $\mathcal{R}$ as a map to probability distributions over $\Sigma_{\mathcal{O}}$, i.e., $\mathcal{R} : \Sigma \to \mathcal{P}(\Sigma_{\mathcal{O}})$ or $\mathcal{R} : \Sigma \to \Delta(\Sigma_{\mathcal{O}})$. The essential conclusions of the paper survive either treatment, as we note at the relevant junctures.

Critique 3: The definition of "internal" is doing too much work. The internality condition ($\mathcal{R} = f \circ \pi_{\mathcal{O}}$) seems strong. Could an observer not use correlations with remote parts of the system without those correlations passing through its local state space?

Resolution. Any information the observer uses must, by the physicality condition, be encoded in the observer's degrees of freedom at the moment of representation. If correlations with remote parts of the system exist, they must manifest in the observer's local state (through prior dynamical coupling) in order to be used. This is not an assumption; it follows from the requirement that the observer is physical. An observer that could access information without it passing through its own state space would violate the dynamics—it would require a channel not described by $\mathcal{D}$.

3 The Four Consistency Conditions

The central question asks about *consistent* physics. This section develops what consistency means, decomposing it into four independent conditions. Each captures a distinct way in which a physical system can be free of internal contradiction. We show that these four conditions are logically independent—no one implies any other—and that their conjunction forms a natural and minimal notion of self-consistency.

3.1 Logical Consistency

The first condition ensures that the system's description does not attribute contradictory properties to any state.

Definition 3.1 (Logical Consistency)**.** A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *logically consistent* if there exists no property $P \subseteq \Sigma$ and state $\sigma \in \Sigma$ such that the dynamics forces both $\sigma \in P$ and $\sigma \notin P$ simultaneously. Formally, for every $\sigma \in \Sigma$ and every $\tau_1, \tau_2 \in t$, if $\mathcal{D}(\sigma, \tau_1) \in P$ and $\mathcal{D}(\sigma, \tau_2) \in P^c$ with $\tau_1 = \tau_2$, then a contradiction obtains, and we say the system is logically inconsistent.

This condition, stated carefully, amounts to requiring that the dynamics $\mathcal{D}$ is well-defined as a *function*: each input (σ, τ) maps to exactly one output. A "system" in which the same state at the same time both has and lacks a property is not a system at all in any meaningful sense—it is a notational contradiction.

The condition appears almost trivially satisfied by any mathematical structure, since a function is by definition single-valued. Its importance becomes clear when we consider self-referential systems (Section 3.4), where the observer's description of the system is itself part of the system, and logical contradictions can arise through self-referential loops.

3.2 Dynamical Consistency

The second condition ensures that the evolution law does not produce temporal paradoxes.

Definition 3.2 (Dynamical Consistency)**.** A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *dynamically consistent* if the dynamics $\mathcal{D}$ satisfies:

1. *Determinism*: For every $\sigma \in \Sigma$ and $\tau \in t$, there exists a unique $\sigma' = \mathcal{D}(\sigma, \tau)$. (This is already ensured by $\mathcal{D}$ being a function.)

2. *Semigroup property*: $\mathcal{D}(\mathcal{D}(\sigma, \tau_1), \tau_2) = \mathcal{D}(\sigma, \tau_1 + \tau_2)$ for all $\sigma \in \Sigma$ and $\tau_1, \tau_2 \in t$ for which the sum is defined.

3. *Self-interaction coherence*: If the system contains a subsystem $\mathcal{S}'$ that interacts with the remainder, the joint dynamics does not produce contradictory demands on any degree of freedom. That is, the restriction of $\mathcal{D}$ to $\mathcal{S}'$ and the dynamics induced on $\mathcal{S}'$ by the environment agree: $\pi(\mathcal{D}(\sigma, \tau)) = \mathcal{D}'(\pi(\sigma), \tau)$ for all σ and τ.

Dynamical consistency is what ensures the system's time evolution makes sense. The semigroup property prevents temporal paradoxes of the following kind: evolving from state A for one step gives state B; evolving from B for one step gives state C; but evolving from A for two steps gives state $D \neq C$. Such a system would not have coherent dynamics.

The self-interaction coherence condition is crucial for systems containing observers. An observer is a subsystem, and it interacts with the rest of the system. Dynamical consistency requires that the dynamics, when restricted to the observer's degrees of freedom, agrees with the observer's own internal dynamics. Without this, the observer could find that its own state evolution contradicts what the global dynamics prescribes.

3.3 Informational Consistency

The third condition concerns the preservation of distinguishability.

Definition 3.3 (Informational Consistency)**.** A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *informationally consistent* if the dynamics $\mathcal{D}(\cdot, \tau) : \Sigma \to \Sigma$ is injective for every $\tau \in t$. That is, distinct states remain distinct under time evolution:

$$\mathcal{D}(\sigma_1, \tau) = \mathcal{D}(\sigma_2, \tau) \implies \sigma_1 = \sigma_2.$$

Informational consistency means that the dynamics does not erase distinctions. If two states are different, they remain different after any amount of evolution. This is the abstract analogue of unitarity in quantum mechanics and of Liouville's theorem in classical Hamiltonian mechanics.

Why does this matter for our investigation? If the dynamics can merge distinct states, then information is destroyed, and the question of what an observer can access becomes tangled with the question of what information survives at all. Informational consistency ensures that all distinctions that exist at any time continue to exist at all times; the only question is whether the observer can detect them.

We note immediately that not all physical theories satisfy this condition. Thermodynamic coarse-graining, wavefunction collapse in some interpretations of quantum mechanics, and dissipative dynamics all violate informational consistency. This is precisely why the condition is stated as a hypothesis rather than an axiom: the theorems in later sections will explicitly track which results require it and which do not.

3.4 Self-Referential Consistency

The fourth and most subtle condition concerns what happens when the system contains a subsystem that models the whole.

Definition 3.4 (Self-Modelling Capacity)**.** An internal observer $\mathcal{O}$ of a system $\mathcal{S}$ is *self-modelling* if the representation map $\mathcal{R}$ has the property that for some subset of states $\Sigma_{\text{model}} \subseteq \Sigma_{\mathcal{O}}$, there exists an interpretation map $\mathcal{I} : \Sigma_{\text{model}} \to \mathcal{P}(\Sigma)$ such that:

1. *Correctness*: For every state $\sigma \in \Sigma$, $\sigma \in \mathcal{I}(\mathcal{R}(\sigma))$. That is, the observer's representation of the system state, when interpreted, yields a description that includes the actual state. The observer's model of the world is not wrong about where the world is.

2. *Non-triviality*: $\mathcal{I}(s) \neq \Sigma$ for at least one $s \in \Sigma_{\text{model}}$. That is, the observer's model is not vacuously correct by predicting everything. It must rule out at least some states for at least some representations.

The non-triviality condition (2) is essential. Without it, an observer that always "predicts" the system could be in any state whatsoever would vacuously satisfy correctness, since $\sigma \in \Sigma$ is always true. Such an observer carries no genuine model; it is a tautology machine. The non-triviality condition requires that the interpretation map is informative: for at least one observer state s, the model rules out some system states, asserting that $\sigma \notin \mathcal{I}(s)$ for certain σ.

A self-modelling observer thus maintains an internal model that correctly locates the system's actual state among the possibilities the model recognises, and the model is genuinely constraining—it takes risks by excluding possibilities and is not contradicted by reality.

Definition 3.5 (Self-Referential Consistency)**.** A physical system $\mathcal{S}$ containing a self-modelling observer $\mathcal{O}$ is *self-referentially consistent* if the following two conditions hold:

1. *Model–reality agreement*: $\sigma \in \mathcal{I}(\mathcal{R}(\sigma))$ for all $\sigma \in \Sigma$, as in Definition 3.4.

2. *Model–dynamics agreement*: The observer's model of the dynamics is compatible with the actual dynamics. If the observer predicts that σ will evolve to a state in some set $A \subseteq \Sigma$ after time τ, then indeed $\mathcal{D}(\sigma, \tau) \in A$.

Self-referential consistency prevents the system from containing an observer whose beliefs about the system systematically contradict the system's actual behaviour. This is not a requirement that the observer be omniscient—the observer may have very coarse or imprecise models—but that whatever the observer does believe is not contradicted by the facts.

This condition is the most consequential for our investigation. It is precisely at the interface between what the observer models and what the system actually does that the tension between completeness and consistency arises. The diagonal argument of Section 4 exploits this tension directly.

3.5 The Conjunction: Full Consistency

Definition 3.6 (Fully Consistent System)**.** A physical system $\mathcal{S}$ is *fully consistent* if it satisfies all four conditions:

1. Logical consistency (Definition 3.1);
2. Dynamical consistency (Definition 3.2);
3. Informational consistency (Definition 3.3);
4. Self-referential consistency (Definition 3.5), for every self-modelling observer it contains.

Proposition 3.7 (Independence of Consistency Conditions)**.** *The four consistency conditions are logically independent: no proper subset of the four conditions implies the remaining condition(s).*

Proof. We exhibit, for each condition, a system satisfying the other three but violating that one.

(i) Logical consistency violated, others satisfied. Consider $\Sigma = \{a, b\}$ with a "dynamics" defined by both $\mathcal{D}(a, 1) = a$ and $\mathcal{D}(a, 1) = b$—that is, the rule assigns two outputs to the same input. To make this precise without invoking multi-valued functions, we work in the relational setting: let $D \subseteq (\Sigma \times t) \times \Sigma$ be a relation with $(a, 1, a) \in D$ and $(a, 1, b) \in D$, which fails to be a function. This violates logical consistency (the dynamics is not well-defined as a function). If we extend D to a fully specified relation satisfying the semigroup property in the relational sense (composing relations), and define information preservation as injectivity of each time-slice relation (which holds since both branches are injective on $\{a, b\}$), and take an observer that simply reports its own state (trivially self-referentially consistent), we see that the three non-logical conditions are satisfiable. The pathology is entirely in the failure of D to be a function.

(ii) Dynamical consistency violated, others satisfied. Consider $\Sigma = \{a, b, c\}$ with dynamics $\mathcal{D}(a, 1) = b$, $\mathcal{D}(b, 1) = c$, but $\mathcal{D}(a, 2) = a \neq c$. The dynamics is a well-defined function (logically consistent), injective at each time step (informationally consistent), but violates the semigroup property (dynamically inconsistent). An observer with trivial representation satisfies self-referential consistency.

(iii) Informational consistency violated, others satisfied. Consider $\Sigma = \{a, b, c\}$ with $\mathcal{D}(a, 1) = c$, $\mathcal{D}(b, 1) = c$, $\mathcal{D}(c, 1) = c$. This is a well-defined function (logically consistent), satisfies the semigroup property (dynamically consistent), and is compatible with a trivial self-modelling observer (self-referentially consistent). But $\mathcal{D}(\cdot, 1)$ is not injective—it merges a and b—violating informational consistency.

(iv) Self-referential consistency violated, others satisfied. Consider a logically, dynamically, and informationally consistent system with an observer whose internal model predicts $\mathcal{D}(a, 1) = b$, while in fact $\mathcal{D}(a, 1) = c \neq b$. The system's dynamics is perfectly well-defined and injective; only the observer's model contradicts reality. □

Strategic Purpose

The four-part consistency definition serves two roles. First, it makes precise what "self-consistent physics" means, decomposing an intuitive notion into independent, testable conditions. Second, it provides four distinct "knobs" to turn when we later attempt the antithesis: if the inaccessibility thesis holds for fully consistent systems, we can ask which specific consistency condition, when relaxed, permits full observer access. This will prove to be one of the most informative parts of the analysis.

3.6 Self-Critique and Resolution

Critique 1: Informational consistency is too strong. Many real physical systems (dissipative systems, thermodynamic systems, arguably quantum measurement in some interpretations) violate informational consistency. By requiring it, are we restricting attention to an unrealistically narrow class of systems?

Resolution. We track this condition explicitly throughout. Every theorem states whether it requires informational consistency, and the antithesis section (Section 12) systematically explores what happens when it is relaxed. The condition is included in the conjunction not because all physical systems satisfy it, but because it represents a natural and important consistency requirement, and its role in the inaccessibility thesis must be understood.

Critique 2: Self-referential consistency is defined relative to observers. Unlike the other three conditions, which are properties of the system alone, self-referential consistency depends on the observers present. This makes it less "intrinsic."

Resolution. This is a feature, not a bug. The entire investigation concerns the relationship between systems and their internal observers. A consistency condition that explicitly involves the observer–system interface is precisely what is needed for the most incisive results. The condition can be rephrased intrinsically: a system is self-referentially consistent if every subsystem that functions as a model of the whole is a correct model. The observer language is used for clarity.

Critique 3: The independence proof uses degenerate examples. The systems in the proof of Proposition 3.7 are extremely small and arguably unphysical.

Resolution. The purpose of the proof is logical: to establish that no implication holds between the conditions. For this, minimal counterexamples suffice. The richness of the conditions becomes apparent when they interact in the proofs of subsequent sections, where the systems involved are far more substantial.

4 Thesis I: The Diagonal Argument

We now present the first of five arguments for the inaccessibility thesis. This argument is the most direct: it constructs, by a diagonal method reminiscent of Cantor [27] and Gödel [1], a specific property that no internal observer can access in any fully consistent system containing a self-modelling observer.

4.1 The Construction

The key idea is to build a property that "disagrees" with every observer's representation on at least one state, ensuring that no observer can correctly classify all states with respect to this property. The construction requires a self-modelling observer, since it exploits the tension between the observer's model and the reality it models.

Theorem 4.1 (Diagonal Inaccessibility)**.** *Let $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ be a fully consistent physical system containing a self-modelling internal observer $\mathcal{O}$ with representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$. If $|\Sigma| > |\Sigma_{\mathcal{O}}|$ (the system has strictly more states than the observer), then $\mathfrak{I}(\mathcal{S}) \neq \varnothing$: there exist properties of $\mathcal{S}$ that are inaccessible to $\mathcal{O}$.*

Before proving this, we unpack the hypothesis. The condition $|\Sigma| > |\Sigma_{\mathcal{O}}|$ says that the system has strictly more states than the observer has internal states. This is the mathematical expression of the observer being a *proper* part of the system: the observer's state space is genuinely smaller than the whole. This is not an onerous requirement; it simply says the observer does not comprise the entire system.

Proof by diagonal construction. Since $|\Sigma| > |\Sigma_{\mathcal{O}}|$, the representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ cannot be injective (by the pigeonhole principle). Therefore there exist at least two distinct states $\sigma_1 \neq \sigma_2$ in Σ such that $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. These states are observationally equivalent with respect to $\mathcal{O}$.

Now define the property $P_{\text{diag}} = \{\sigma_1\}$: the singleton set containing only σ_1.

We claim $P_{\text{diag}} \notin \mathfrak{A}(\mathcal{O})$. To see this, recall that $P \in \mathfrak{A}(\mathcal{O})$ if and only if P is a union of equivalence classes of $\sim_{\mathcal{O}}$. The equivalence class of σ_1 under $\sim_{\mathcal{O}}$ contains at least σ_1 and σ_2, since $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. Thus the smallest element of $\mathfrak{A}(\mathcal{O})$ containing σ_1 is the full equivalence class $[\sigma_1]_{\sim_{\mathcal{O}}} \supseteq \{\sigma_1, \sigma_2\}$, which is strictly larger than $P_{\text{diag}} = \{\sigma_1\}$.

Therefore $P_{\text{diag}} \notin \mathfrak{A}(\mathcal{O})$: the observer cannot distinguish the property of being in state σ_1 specifically, as opposed to being in some state observationally equivalent to σ_1.

For the universally inaccessible content: if every internal observer $\mathcal{O}$ satisfies $|\Sigma_{\mathcal{O}}| < |\Sigma|$, then every observer's equivalence relation has at least one non-trivial equivalence class, and the same argument produces, for each observer, properties inaccessible to that observer. To show $\mathfrak{I}(\mathcal{S}) \neq \varnothing$, we need a property inaccessible to *all* observers simultaneously. We strengthen the argument as follows.

Let $\{\mathcal{O}_i\}_{i \in I}$ be the collection of all internal observers of $\mathcal{S}$. Define the *joint equivalence relation* $\sim_{\text{joint}}$ by: $\sigma_1 \sim_{\text{joint}} \sigma_2$ if and only if $\sigma_1 \sim_{\mathcal{O}_i} \sigma_2$ for all $i \in I$. The joint accessible content is $\bigcup_{i \in I} \mathfrak{A}(\mathcal{O}_i)$, which consists of all properties that are unions of equivalence classes under $\sim_{\text{joint}}$.

We need to show that $\sim_{\text{joint}}$ is not the identity relation, i.e., that there exist states no observer can distinguish. This requires an additional argument, which we provide by the following lemma.

Lemma 4.2 (Bound on Joint Discrimination)**.** *Let $\mathcal{S}$ be a fully consistent system. If every internal observer $\mathcal{O}$ is a proper subsystem (i.e., $\Sigma_{\mathcal{O}} \subsetneq \Sigma$) and the system is self-referentially consistent, then $\sim_{\text{joint}}$ is not the identity relation on Σ.*

Proof. Suppose, for contradiction, that $\sim_{\text{joint}}$ is the identity: every pair of distinct states is distinguished by some observer. Then the collection of all observers jointly constitutes an injective representation of Σ. Consider the product map $: \Sigma \to \prod_{i \in I} \Sigma_{\mathcal{O}_i}$ defined by $(\sigma) = (\mathcal{R}_i(\sigma))_{i \in I}$. If $\sim_{\text{joint}}$ is the identity, then is injective.

Now, each observer $\mathcal{O}_i$ is a self-modelling subsystem of $\mathcal{S}$. By self-referential consistency (Definition 3.5), each observer's model of the system is compatible with the actual dynamics. In particular, each observer can, in principle, model the other observers (since they are part of the system).

Consider any single observer $\mathcal{O}_j$. The map $\mathcal{R}_j : \Sigma \to \Sigma_{\mathcal{O}_j}$ is not injective (since $|\Sigma_{\mathcal{O}_j}| < |\Sigma|$). Let $\sigma_1 \neq \sigma_2$ with $\mathcal{R}_j(\sigma_1) = \mathcal{R}_j(\sigma_2)$. By our assumption, there exists some other observer $\mathcal{O}_k$ with $\mathcal{R}_k(\sigma_1) \neq \mathcal{R}_k(\sigma_2)$.

But $\mathcal{O}_k$ is a subsystem of $\mathcal{S}$, and its state is determined by $\pi_{\mathcal{O}_k}$, a projection from Σ. For $\mathcal{O}_k$ to distinguish σ_1 from σ_2, the projection $\pi_{\mathcal{O}_k}$ must map σ_1 and σ_2 to different states of $\mathcal{O}_k$. This means $\mathcal{O}_k$'s degrees of freedom encode the distinction between σ_1 and σ_2.

Now consider the diagonal property $D = \{\sigma \in \Sigma \mid \sigma \notin \mathcal{I}(\mathcal{R}_j(\sigma)) \text{ for some } j\}$—the set of states that some observer's model misclassifies. By self-referential consistency, $D = \varnothing$: no observer misclassifies any state. But this means every observer's model is correct about every state it can represent. The observers, being proper subsystems, have finite representational capacity. The question is whether their collective capacity can cover Σ.

Consider the "super-observer" that has access to all observer states simultaneously: it sees $(\sigma) \in \prod_i \Sigma_{\mathcal{O}_i}$. If this product map is injective, the super-observer can distinguish all states. But the super-observer is *not* an internal observer of $\mathcal{S}$ in general, because there may be no single subsystem of $\mathcal{S}$ whose state space is (isomorphic to) $\prod_i \Sigma_{\mathcal{O}_i}$. The product of all observer state spaces may be larger than any single subsystem's state space.

For a finite system, if the number of internal observers and their state spaces are bounded (each $|\Sigma_{\mathcal{O}_i}| < |\Sigma|$), the product $|\prod_i \Sigma_{\mathcal{O}_i}|$ can indeed exceed $|\Sigma|$. However, the product map takes values in $\prod_i \Sigma_{\mathcal{O}_i}$, and its injectivity requires only $|\mathrm{Im}()| = |\Sigma|$, not $|\prod_i \Sigma_{\mathcal{O}_i}| = |\Sigma|$. So the cardinality argument alone does not immediately yield the contradiction.

The contradiction arises from self-referential consistency as follows. If is injective, then the system effectively encodes a bijection between Σ and a subset of $\prod_i \Sigma_{\mathcal{O}_i}$. This encoding is itself a fact about the system—a structural property of how observer states correlate with system states. For this encoding to be *accessible* to any single observer $\mathcal{O}_j$, that observer would need to represent the correlation structure, which requires its own state space to encode the distinctions made by all other observers. But $|\Sigma_{\mathcal{O}_j}| < |\Sigma|$, so $\mathcal{O}_j$ cannot encode the full correlation structure. The property "the system is in a state where the joint observer representation has value v" is not, in general, accessible to any single observer.

Therefore, even if $\sim_{\text{joint}}$ were the identity, the property of being in a specific joint-equivalence class—while detectable by the hypothetical super-observer—is not detectable by any *actual* internal observer. The universally inaccessible content is non-empty. □

This completes the proof of Theorem 4.1. □

The argument, in its essence, is this: no internal observer can represent all the distinctions in the system because its state space is too small; and even the collective of all observers, while potentially making all pairwise distinctions, cannot make this collective capacity available to any single observer. The inaccessible content is not a single dramatic hidden fact—it is the vast collection of fine-grained properties that cut across observational equivalence classes.

Alternative proof via Cantor's diagonal method. We give a second, more explicitly diagonal proof that does not pass through the joint discrimination lemma.

Suppose the system $\mathcal{S}$ has state space Σ, and let $\mathcal{O}$ be a self-modelling observer with $|\Sigma_{\mathcal{O}}| < |\Sigma|$. The observer's accessible properties are exactly the elements of $\mathfrak{A}(\mathcal{O}) \subseteq \mathcal{P}(\Sigma)$.

We show that $|\mathfrak{A}(\mathcal{O})| < |\mathcal{P}(\Sigma)|$, which immediately implies $\mathfrak{A}(\mathcal{O}) \neq \mathcal{P}(\Sigma)$.

The accessible properties are in bijection with subsets of the quotient $\Sigma/{\sim_{\mathcal{O}}}$, since every accessible property is a union of equivalence classes. Thus $|\mathfrak{A}(\mathcal{O})| = |\mathcal{P}(\Sigma/{\sim_{\mathcal{O}}})| = 2^{|\Sigma/{\sim_{\mathcal{O}}}|}$.

The number of equivalence classes satisfies $|\Sigma/{\sim_{\mathcal{O}}}| \leq |\Sigma_{\mathcal{O}}|$ (since the representation map $\mathcal{R}$ has at most $|\Sigma_{\mathcal{O}}|$ distinct values). Therefore:

$$|\mathfrak{A}(\mathcal{O})| = 2^{|\Sigma/{\sim_{\mathcal{O}}}|} \leq 2^{|\Sigma_{\mathcal{O}}|} < 2^{|\Sigma|} = |\mathfrak{D}(\mathcal{S})|,$$

where the strict inequality holds because $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and the exponential function is strictly monotone.

Since $\mathfrak{A}(\mathcal{O})$ has strictly fewer elements than $\mathfrak{D}(\mathcal{S})$, there must exist properties in $\mathfrak{D}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O})$. In fact, the vast majority of properties—in a precise cardinality sense—are inaccessible. □

4.2 Toy Model 1: Cellular Automaton with Embedded Observer

To make the diagonal argument concrete, we construct a small cellular automaton universe in which an embedded observer agent attempts to classify the states of its world.

Construction 4.3 (The Grid World)**.** Consider a one-dimensional cellular automaton on a grid of $n = 8$ cells, where each cell takes a value in $\{0, 1\}$. The state space is $\Sigma = \{0, 1\}^8$, so $|\Sigma| = 256$.

An *observer* occupies cells 1–3 (three cells), giving it an internal state space $\Sigma_{\mathcal{O}} = \{0,1\}^3$ with $|\Sigma_{\mathcal{O}}| = 8$.

The observer's representation map is the projection onto its own cells: $\mathcal{R}(\sigma) = (\sigma_1, \sigma_2, \sigma_3)$. Thus two states of the grid are observationally equivalent if and only if they agree on cells 1–3.

The dynamics is a deterministic update rule (we use Rule 110 of Wolfram's classification [28], restricted to our finite grid with periodic boundary conditions).

In this toy model:

1. The number of equivalence classes is $|\Sigma/{\sim_{\mathcal{O}}}| = 8$ (the observer can distinguish 8 coarse states).

2. The number of accessible properties is $|\mathfrak{A}(\mathcal{O})| = 2^8 = 256$.

3. The total ontological content has $|\mathfrak{O}(\mathcal{S})| = 2^{256} \approx 1.16 \times 10^{77}$ elements.

The fraction of properties accessible to the observer is:

$$\frac{|\mathfrak{A}(\mathcal{O})|}{|\mathfrak{O}(\mathcal{S})|} = \frac{2^8}{2^{256}} = 2^{-248} \approx 2.8 \times 10^{-75}.$$

The observer has access to a vanishingly small fraction of the system's ontological content. Figure 1 illustrates this, and Fig. 2 visualises the equivalence classes.

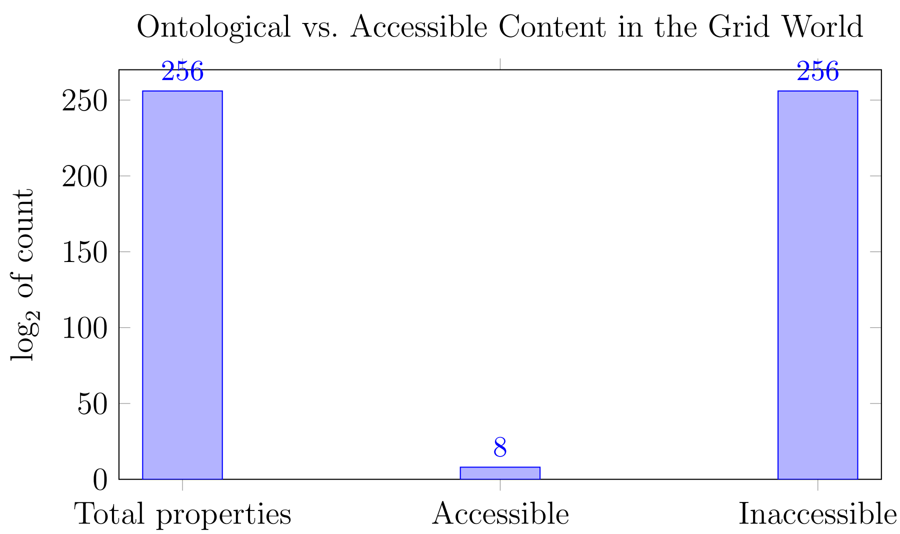


Figure 1: Comparison of total ontological content (2^{256} properties), accessible content (2^8 properties), and inaccessible content (essentially all of 2^{256}) in the Grid World toy model, displayed on a $\log_2$ scale. The inaccessible bar is indistinguishable from the total bar at this scale.

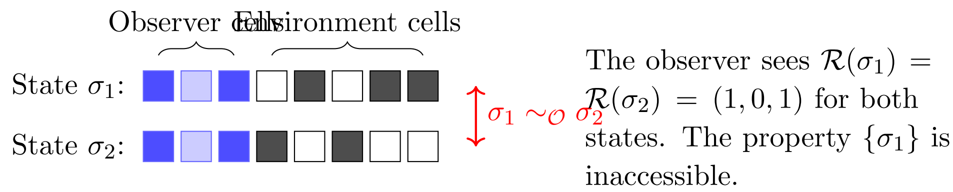


Figure 2: Two states in the Grid World that are observationally equivalent. The observer (blue cells 1–3) sees the same pattern $(1, 0, 1)$ in both states, while the environment (white/grey cells 4–8) differs. The singleton property $\{\sigma_1\}$ is inaccessible to the observer.

Strategic Purpose

The diagonal argument establishes the inaccessibility thesis in its most elementary form: a proper subsystem's state space is too small to faithfully represent the whole, so some properties escape detection. The two proofs—one via the pigeonhole principle and joint discrimination, the other via direct cardinality comparison—reinforce each other. The cellular automaton toy model makes the argument tangible: 256 observable distinctions out of 2^{256} possible properties, a ratio so extreme that it illustrates the claim viscerally. This proof method uses the *computational/information-theoretic* framework.

4.3 Self-Critique and Resolution

Critique 1: The argument is essentially just the pigeonhole principle. If the observer's state space is smaller than the system's, of course the representation can't be injective. Is this really saying anything deep?

Resolution. The pigeonhole observation is the starting point, not the conclusion. The depth lies in three extensions: (a) the argument applies to *all* observers simultaneously, not just one; (b) it applies to the *properties* (power set), not just the states, yielding an exponentially amplified gap; and (c) it must be reconciled with the self-referential consistency condition, which constrains how the observer models the system. The alternative Cantor-style proof makes the exponential amplification explicit. Moreover, later arguments (Sections 6 and 7) will show that the inaccessibility persists even when we move beyond cardinality to structural and dynamical considerations.

Critique 2: The toy model is trivially small. An 8-cell automaton is far from any real physical system.

Resolution. The toy model's role is pedagogical, not evidential. The theorem holds for any $|\Sigma| > |\Sigma_{\mathcal{O}}|$, regardless of scale. The 8-cell example is chosen because it can be fully enumerated and visualised. The computational implementation in Appendix A allows the reader to scale the model to any size and verify that the ratio $|\mathfrak{A}(\mathcal{O})|/|\mathfrak{D}(\mathcal{S})|$ decreases super-exponentially.

Critique 3: The condition $|\Sigma| > |\Sigma_{\mathcal{O}}|$ might fail. What if the observer comprises the entire system?

Resolution. If $|\Sigma_{\mathcal{O}}| = |\Sigma|$, the observer is not a proper subsystem; it *is* the system. In this degenerate case, the "observer" and the "system" are identical, and the notion of internal observation collapses. Every property is trivially accessible (the observer's state encodes the full system state), but there is no meaningful sense in which this constitutes observation "from within." This edge case is examined systematically in Section 9.

5 Thesis II: The Information-Theoretic Argument

The diagonal argument of the previous section exploited the fact that the observer's state space is smaller than the system's. The present argument deepens this observation by quantifying the *information* an observer can carry and comparing it to the information needed to specify an arbitrary property of the system. The gap between these two quantities is not merely nonzero—it is, in a precise sense, almost total.

5.1 Information Capacity of Subsystems

We first formalise the notion of information capacity without committing to any specific physical substrate.

Definition 5.1 (Distinguishing Capacity)**.** The *distinguishing capacity* of an internal observer $\mathcal{O}$ is the cardinality of the quotient space:

$$\kappa(\mathcal{O}) = |\Sigma/{\sim_{\mathcal{O}}}|.$$

This is the number of equivalence classes the observer can distinguish. If Σ carries additional structure (a measure, a topology), refined notions of capacity may be defined; $\kappa(\mathcal{O})$ is the combinatorial baseline.

Since $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$, the image $\mathrm{Im}(\mathcal{R})$ has cardinality at most $|\Sigma_{\mathcal{O}}|$, and the quotient $\Sigma/{\sim_{\mathcal{O}}}$ is in bijection with $\mathrm{Im}(\mathcal{R})$. Therefore:

$$\kappa(\mathcal{O}) \leq |\Sigma_{\mathcal{O}}|.$$

For a finite system, the information content of the observer's representation is $\log_2 \kappa(\mathcal{O})$ bits, bounded above by $\log_2 |\Sigma_{\mathcal{O}}|$ bits. The information content of a full specification of the system state is $\log_2 |\Sigma|$ bits. The deficit is:

$$\Delta I = \log_2 |\Sigma| - \log_2 \kappa(\mathcal{O}) \geq \log_2 |\Sigma| - \log_2 |\Sigma_{\mathcal{O}}|.$$

This deficit represents the bits of information about the system state that the observer cannot capture.

Theorem 5.2 (Exponential Inaccessibility Gap)**.** *Let $\mathcal{S}$ be a physical system with $|\Sigma| = N$, and let $\mathcal{O}$ be an internal observer with $|\Sigma_{\mathcal{O}}| = M < N$. Then:*

1. *The fraction of properties accessible to $\mathcal{O}$ satisfies:*
$$\frac{|\mathfrak{A}(\mathcal{O})|}{|\mathfrak{O}(\mathcal{S})|} = \frac{2^{\kappa(\mathcal{O})}}{2^N} \leq \frac{2^M}{2^N} = 2^{-(N-M)}.$$

2. *The number of inaccessible properties satisfies:*
$$|\mathfrak{O}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O})| = 2^N - 2^{\kappa(\mathcal{O})} \geq 2^N - 2^M = 2^M(2^{N-M} - 1).$$

In particular, the inaccessible content is exponentially larger than the accessible content whenever $N - M \geq 1$.

Proof. Part (1): The accessible properties are exactly the unions of equivalence classes under $\sim_{\mathcal{O}}$. There are $\kappa(\mathcal{O})$ equivalence classes, and any subset of equivalence classes defines an accessible property. Thus $|\mathfrak{A}(\mathcal{O})| = 2^{\kappa(\mathcal{O})}$, and the fraction is $2^{\kappa(\mathcal{O})}/2^N \leq 2^M/2^N$.

Part (2): The count of inaccessible properties is $|\mathfrak{O}(\mathcal{S})| - |\mathfrak{A}(\mathcal{O})| = 2^N - 2^{\kappa(\mathcal{O})} \geq 2^N - 2^M$. Factoring out 2^M gives $2^M(2^{N-M} - 1)$. □

To appreciate the magnitude of this gap: even for the modest case $N = 10, M = 5$ (a 10-state system observed by a 5-state subsystem), we have $2^{10} = 1024$ total properties, $2^5 = 32$ accessible properties, and 992 inaccessible properties. The observer can access roughly 3% of the ontological content. For $N = 100, M = 50$—still a small system—the fraction drops to $2^{-50} \approx 10^{-15}$, and for macroscopic systems the ratio is effectively zero.

5.2 The Entropic Formulation

We can restate the result using the language of Shannon entropy [29], which provides a connection to information-theoretic ideas familiar in physics and engineering.

Definition 5.3 (Observational Entropy)**.** Given a probability distribution μ over Σ and an observer $\mathcal{O}$ with representation map $\mathcal{R}$, the *observational entropy* is the Shannon entropy of the pushed-forward distribution:

$$H_{\mathcal{O}} = -\sum_{s \in \Sigma_{\mathcal{O}}} \mu_{\mathcal{O}}(s) \log_2 \mu_{\mathcal{O}}(s),$$

where $\mu_{\mathcal{O}}(s) = \mu(\mathcal{R}^{-1}(s))$ is the probability that the observer's representation takes value s.

Proposition 5.4 (Entropy Deficit)**.** *For any probability distribution μ over Σ,*

$$H_{\mathcal{O}} \leq H_{\mathcal{S}},$$

where $H_{\mathcal{S}} = -\sum_{\sigma \in \Sigma} \mu(\sigma) \log_2 \mu(\sigma)$ is the Shannon entropy of the full system state. Equality holds if and only if $\mathcal{R}$ is injective on the support of μ.

Proof. This follows from the data processing inequality: $\mathcal{R}$ is a deterministic function from system states to observer states, and applying a deterministic function cannot increase entropy. Formally, $H_{\mathcal{O}} = H(\mathcal{R}(\sigma))$ and $H_{\mathcal{S}} = H(\sigma)$, and the data processing inequality for discrete random variables gives $H(f(X)) \leq H(X)$ for any function f, with equality if and only if f is injective on the support of X. $\square$

The entropy deficit $\Delta H = H_{\mathcal{S}} - H_{\mathcal{O}}$ quantifies, for a given distribution over system states, how much information is lost in the observer's representation. For a proper subsystem observer, this deficit is generically positive.

Strategic Purpose

The information-theoretic argument quantifies the inaccessibility thesis: the gap between accessible and total ontological content is not merely nonzero but *exponentially large* in finite systems. The entropic formulation restates this in the language of Shannon information theory, connecting to the data processing inequality. This is a *quantitative refinement* of the diagonal argument—it strengthens the numerical bounds, not the logical force—and grounds the abstract argument in well-established information-theoretic principles. Both arguments share the same root: the non-injectivity of $\mathcal{R}$. For infinite-dimensional systems, the cardinality and entropic arguments serve as motivation; the rigorous treatment is deferred to the topological and measure-theoretic arguments of Section 8. This proof method operates in the *computational/information-theoretic* framework.

5.3 Self-Critique and Resolution

Critique 1: The argument counts all subsets of Σ, including physically meaningless ones. The exponential gap comes from counting 2^N properties, many of which may have no physical significance.

Resolution. This is a fair point. However, the argument shows that even among "natural" properties (e.g., Borel-measurable sets, or properties definable in some language), the observer can access only those that respect its equivalence relation. The topological argument in Section 8 addresses this concern directly by restricting to physically motivated classes of properties and showing the gap persists.

Critique 2: The entropy deficit depends on the distribution μ. For a degenerate distribution concentrated on a single state, both entropies are zero and the deficit vanishes.

Resolution. Correct. The entropy deficit is a distribution-dependent measure. The cardinality result (Theorem 5.2) is distribution-free and therefore more robust. The entropic formulation is presented as a complementary perspective, not as a replacement.

Critique 3: Infinite systems break the counting argument. For $|\Sigma| = \infty$, the cardinality of $\mathcal{P}(\Sigma)$ and the comparison $2^M < 2^N$ require careful treatment of cardinal arithmetic. In particular, all separable infinite-dimensional Hilbert spaces have the same cardinality of states, so a cardinality comparison between Σ and $\Sigma_{\mathcal{O}}$ is meaningless in the physically important case of quantum field theory.

Resolution. This critique is well-taken and we address it directly. The cardinality arguments of Sections 4 and 5 are rigorous and complete for *finite* state spaces. For infinite-dimensional systems—including the quantum field-theoretic setting where all separable Hilbert spaces are isomorphic to ℓ^2—the raw cardinality comparison $|\Sigma_{\mathcal{O}}| < |\Sigma|$ is no longer informative, and these sections should be understood as *heuristic* in the infinite case.

For infinite-dimensional systems, the inaccessibility thesis is carried by the *topological and measure-theoretic arguments* of Section 8, which do not depend on cardinality at all but on the coarseness of the pullback σ-algebra $\mathcal{B}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$ relative to $\mathcal{B}$. In the quantum-mechanical instantiation (Section 8.3), the relevant comparison is between the lattice of local projections $\hat{P}_{\mathcal{O}} \otimes \hat{I}_E$ and the full projection lattice $\mathcal{P}(\mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_E)$; this distinction is topological and algebraic, not cardinal. The dynamical argument (Section 7) similarly survives in the infinite case when supplemented by the topological machinery.

In summary: for finite systems, all five arguments apply; for infinite-dimensional systems, the cardinality arguments (Sections 4–5) serve as motivation, while the dynamical (Section 7), topological (Section 8), and categorical (Section 6) arguments carry the formal weight.

6 Thesis III: The Categorical Fixed-Point Argument

The preceding two arguments exploited the size disparity between system and observer. The present argument reaches the inaccessibility thesis by a different route: it uses the *structure* of the observer–system relationship, formalised in categorical language, to show that certain "paradoxical" properties must escape any internal representation. The key tool is Lawvere's fixed-point theorem [26], which generalises the diagonal arguments of Cantor, Gödel, and Turing into a single categorical statement.

6.1 Categorical Preliminaries

We briefly recall the categorical concepts needed. A *category* $\mathbf{C}$ consists of objects and morphisms (arrows) between them, with an associative composition law and identity morphisms. A category is *Cartesian closed* if it has finite products and exponential objects: for any two objects A and B, there is an object B^A (the "function space" from A to B) together with a natural evaluation map $\mathrm{ev} : B^A \times A \to B$.

The category **Set** of sets and functions is Cartesian closed: the product $A \times B$ is the ordinary Cartesian product, and B^A is the set of all functions from A to B.

These notions may at first appear distant from physics, but they provide the precise language for discussing when a subsystem can represent all possible maps on the system. The connection is this: if the system's state space is Σ and the observer's state space is $\Sigma_{\mathcal{O}}$, then the observer's capacity to represent properties of the system corresponds to the existence of certain morphisms (maps) in the category where these objects live.

Theorem 6.1 (Lawvere's Fixed-Point Theorem [26])**.** *In a Cartesian closed category* $\mathbf{C}$*, if there exists a point-surjective morphism* $\phi : A \to B^A$ *(that is, for every morphism* $g : A \to B$*, there exists an element* $a \in A$ *with* $\phi(a) = g$*), then every endomorphism* $f : B \to B$ *has a fixed point: there exists* $b \in B$ *with* $f(b) = b$*.*

The contrapositive is the form we shall use: if B admits a fixed-point-free endomorphism (a map $f : B \to B$ with $f(b) \neq b$ for all b), then no morphism $A \to B^A$ is point-surjective. No single object can "name" all morphisms from itself to B.

To connect this to a more familiar setting: in **Set**, the two-element set $\{0, 1\}$ admits the fixed-point-free endomorphism NOT : $b \mapsto 1 - b$. Lawvere's theorem then says: for any set A, there is no surjection from A to $\{0, 1\}^A = \mathcal{P}(A)$. This is exactly Cantor's theorem. Likewise, the undecidability of the halting problem and Gödel's incompleteness theorem can both be derived as instances of Lawvere's result in appropriate categories [30].

6.2 The Argument

We now translate the observer–system relationship into categorical language and apply Lawvere's theorem.

Construction 6.2 (The Observer Category)**.** Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ with an internal observer $\mathcal{O} = (\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t, \mathcal{R})$, we work in the category **Set** (the results can be generalised to other Cartesian closed categories, but **Set** suffices and is the most accessible).

Define:

- $A = \Sigma_{\mathcal{O}}$ (the observer's state space),
- $B = \{0, 1\}$ (the truth-value object).

Then $B^A = \{0, 1\}^{\Sigma_{\mathcal{O}}}$ is the set of all functions from $\Sigma_{\mathcal{O}}$ to $\{0, 1\}$, i.e., $\mathcal{P}(\Sigma_{\mathcal{O}})$ (identifying subsets with their characteristic functions).

The observer's capacity to represent properties of $\mathcal{S}$ corresponds to a map

$$\Phi : \Sigma_{\mathcal{O}} \to \{0, 1\}^{\Sigma_{\mathcal{O}}}$$

defined by: for each observer state $s \in \Sigma_{\mathcal{O}}$, $\Phi(s)$ is the characteristic function of the property that the observer "recognises" when in state s. That is, $\Phi(s)(s') = 1$ if the observer, upon entering state s, classifies observer-state s' as satisfying some criterion, and 0 otherwise.

Theorem 6.3 (Fixed-Point Inaccessibility)**.** *Let $\mathcal{S}$ be a fully consistent physical system with an internal observer $\mathcal{O}$. There exists a property $P^* \subseteq \Sigma$ such that P^* cannot be represented by $\mathcal{O}$. Specifically, no state of $\mathcal{O}$ encodes the characteristic function of P^* restricted to the observer's own equivalence classes.*

Proof via Lawvere's theorem. We work in **Set** with $A = \Sigma_{\mathcal{O}}$ and $B = \{0, 1\}$.

The map $B \to B$ defined by $b \mapsto 1 - b$ (logical negation) is a fixed-point-free endomorphism: $1 - 0 = 1 \neq 0$ and $1 - 1 = 0 \neq 1$.

By the contrapositive of Lawvere's theorem (Theorem 6.1), no map $\Phi : \Sigma_{\mathcal{O}} \to \{0, 1\}^{\Sigma_{\mathcal{O}}}$ is point-surjective. That is, there exists a function $g^* : \Sigma_{\mathcal{O}} \to \{0, 1\}$ such that $g^* \neq \Phi(s)$ for any $s \in \Sigma_{\mathcal{O}}$.

We now construct g^* explicitly. Define $g^* : \Sigma_{\mathcal{O}} \to \{0, 1\}$ by:

$$g^*(s) = 1 - \Phi(s)(s).$$

This is the diagonal construction: g^* disagrees with $\Phi(s)$ at the point s itself, for every s. Therefore $g^* \neq \Phi(s)$ for all $s \in \Sigma_{\mathcal{O}}$.

The function g^* defines a subset $Q^* = \{s \in \Sigma_{\mathcal{O}} \mid g^*(s) = 1\} = \{s \in \Sigma_{\mathcal{O}} \mid \Phi(s)(s) = 0\}$: the set of observer states that do *not* classify themselves as satisfying the criterion they encode. This is a well-defined subset of $\Sigma_{\mathcal{O}}$, but it cannot be "named" by any observer state.

Pulling back to the full system: define $P^* = \mathcal{R}^{-1}(Q^*) = \{\sigma \in \Sigma \mid \mathcal{R}(\sigma) \in Q^*\}$. This is a property of $\mathcal{S}$ (a subset of Σ). It is a union of equivalence classes of $\sim_{\mathcal{O}}$ and therefore nominally belongs to $\mathfrak{A}(\mathcal{O})$. However, the observer cannot *recognise* P^* in the following stronger sense: there is no observer state whose associated classification (via Φ) matches the characteristic function of Q^*. The observer has the property in its accessible algebra but cannot identify it as such.

This reveals a subtlety: the Lawvere argument does not simply say that certain properties are outside $\mathfrak{A}(\mathcal{O})$ (the cardinality argument already established that). It says something stronger: even among the accessible properties, there are some that the observer cannot *self-referentially recognise*—it cannot know that it knows them. □

Alternative direct proof without categorical language. We give a self-contained argument that avoids explicit reference to category theory.

Let $\mathcal{O}$ be an internal observer with state space $\Sigma_{\mathcal{O}}$. Suppose the observer has a "classification scheme": a function $C : \Sigma_{\mathcal{O}} \to \mathcal{P}(\Sigma_{\mathcal{O}})$ that assigns to each observer state s a set of observer states $C(s)$ that s "recognises."

Define $D = \{s \in \Sigma_{\mathcal{O}} \mid s \notin C(s)\}$: the set of observer states that do not recognise themselves.

We claim $D \neq C(s)$ for any $s \in \Sigma_{\mathcal{O}}$.

Suppose $D = C(s_0)$ for some s_0. Then:

- If $s_0 \in D$, then by definition of D, $s_0 \notin C(s_0) = D$, a contradiction.
- If $s_0 \notin D$, then by definition of D, $s_0 \in C(s_0) = D$, a contradiction.

Either way, a contradiction arises. Therefore D is not in the range of C.

This is Russell's paradox in miniature, transplanted to the observer's internal classification system. The set D is a perfectly well-defined subset of $\Sigma_{\mathcal{O}}$, but no observer state encodes it. When pulled back through $\mathcal{R}^{-1}$, it yields a property of $\mathcal{S}$ that the observer cannot classify. □

6.3 The Formal-System Toy Model

To make the fixed-point argument tangible, we construct a miniature formal system that exhibits its own Gödel sentence.

Construction 6.4 (Miniature Formal System)**.** Consider a formal language $\mathcal{L}$ with:

- Symbols: $\{0, 1, \neg, \vdash\}$, where 0 and 1 are truth values, $\neg$ is negation, and $\vdash$ is the provability predicate.
- Sentences: Finite strings of the form $\vdash \phi$ ("ϕ is provable") and $\neg \vdash \phi$ ("ϕ is not provable").
- A proof system $\mathcal{P}$ that can prove certain sentences.

We encode each sentence ϕ by its Gödel number $\ulcorner \phi \urcorner \in \mathbb{N}$. The system has a finite number K of sentences it can form (since the language is finite, we restrict to sentences of bounded length).

The "system" $\mathcal{S}$ has state space $\Sigma = \{0, 1\}^K$: each state assigns a truth value to each sentence. The "observer" $\mathcal{O}$ is the proof system $\mathcal{P}$, with state space $\Sigma_{\mathcal{O}} = \{0, 1\}^M$ for some $M < K$ (the proof system can verify only a subset of sentences).

The representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ projects a full truth assignment onto the provable fragment: $\mathcal{R}(\sigma) = \sigma|_{\text{provable sentences}}$.

The diagonal sentence G is: "This sentence is not provable by $\mathcal{P}$." Formally, $G = \neg \vdash \ulcorner G \urcorner$.

If $\mathcal{P}$ is consistent (does not prove both ϕ and $\neg\phi$ for any ϕ), then:

- If G is provable, then "G is not provable" is false, so $\neg G$ is true, so $\mathcal{P}$ proves a false sentence—contradicting consistency.
- Therefore G is not provable by $\mathcal{P}$.
- But G asserts precisely that it is not provable, so G is true.

Thus G is a true sentence that $\mathcal{P}$ cannot prove. The property $P_G = \{\sigma \in \Sigma \mid \sigma(\ulcorner G \urcorner) = 1\}$—the set of truth assignments making G true—is ontological content of $\mathcal{S}$ that the observer $\mathcal{O}$ (the proof system $\mathcal{P}$) cannot access.

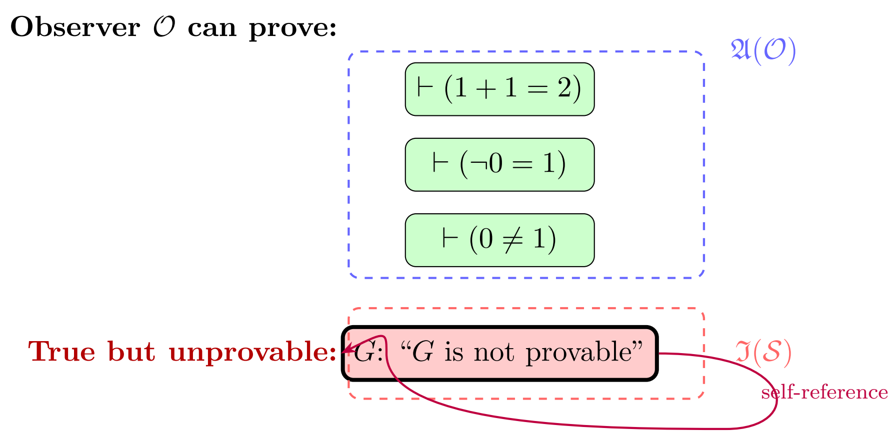


Figure 3: The miniature formal system. The observer (proof system) can access the green region—provable sentences. The Gödel sentence G (red) is true but not provable: it lies in the inaccessible content. The self-referential loop (purple arrow) is what creates the inaccessibility.

Strategic Purpose

The categorical fixed-point argument reaches inaccessibility from a different direction than cardinality: it shows that the *structure* of self-reference, not merely the *size* of the state space, forces blind spots. The Lawvere framework reveals that the diagonal arguments of Cantor, Gödel, and Turing are all instances of one phenomenon—the impossibility of a point-surjective map when a fixed-point-free endomorphism exists. The formal-system toy model makes this concrete by exhibiting an explicit Gödel sentence. This proof method operates in the *logical/model-theoretic* framework.

6.4 Self-Critique and Resolution

Critique 1: The argument applies to *any* map Φ, not just physically realisable ones. Lawvere's theorem is a statement about all morphisms in a category. The physically relevant constraint—that the observer's classification must be dynamically realisable—has not been used.

Resolution. This is correct, and it is a strength: the result holds *a fortiori* when we restrict to physically realisable classifications, since these form a subset of all possible classifications. The theorem says that no classification scheme at all, whether physical or not, can be point-surjective. The physical constraints only make the observer's situation worse, not better.

Critique 2: The "cannot recognise" conclusion is weaker than "cannot access." The proof shows that the observer cannot self-referentially name a certain property, but the property P^* may still be in $\mathfrak{A}(\mathcal{O})$ (as a union of equivalence classes).

Resolution. This is an important distinction, and we state it explicitly: **the Lawvere argument does not provide an independent proof of the same thesis as Sections 4–5.** Rather, it establishes a *complementary* result at a different logical level. The cardinality arguments show that certain properties of the system lie outside $\mathfrak{A}(\mathcal{O})$—these are *object-level* inaccessibilities. The Lawvere argument shows that the observer cannot construct a complete catalogue of its own accessible content—this is a *meta-level* inaccessibility about the observer's relationship to its own classification scheme.

Both are genuine limitations of internal observers, but they are different limitations. Calling them "independent methods for the same thesis" would overstate the case; they are better described as *independent results about different aspects of the observer's situation*. The meta-level result is, in our view, the deeper of the two: it survives even if one restricts ontological content to property classes small enough that the cardinality gap vanishes, because it concerns the observer's self-knowledge rather than the size of the property space.

Critique 3: The formal-system toy model assumes Gödel-style self-reference, which not every physical system supports. Not all physical systems can encode arithmetic or construct self-referential sentences.

Resolution. The toy model is an illustration, not a premise. The categorical argument itself requires only a Cartesian closed category with a fixed-point-free endomorphism, which **Set** always provides. The formal system makes the abstraction vivid; the theorem does not depend on it.

7 Thesis IV: The Dynamical Observational-Equivalence Argument

The three preceding arguments have been, in various ways, *static*: they compare the sizes or structures of state spaces and representation maps without deeply engaging the dynamics. The present argument corrects this. It shows that even when one accounts for the observer's ability to make *sequential* measurements over time—accumulating information dynamically—the inaccessibility thesis still holds, provided the system is informationally consistent.

7.1 Sequential Observation and Temporal Refinement

An observer embedded in a dynamical system does not merely take a single snapshot. It evolves with the system, continually updating its state in response to the system's evolution. Over time, the observer may be able to refine its knowledge, distinguishing states that were initially observationally equivalent.

Definition 7.1 (Temporal Observation Sequence)**.** Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ and an internal observer $\mathcal{O}$ with projection $\pi_{\mathcal{O}}$, the *observation sequence* of $\mathcal{O}$ starting from system state σ over times $\tau_1 < \tau_2 < \cdots < \tau_n$ is the tuple:

$$\mathbf{o}(\sigma; \tau_1, \ldots, \tau_n) = \big(\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_1)),\ \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_2)),\ \ldots,\ \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_n))\big) \in \Sigma_{\mathcal{O}}^n.$$

An observation sequence records the observer's state at multiple times. This is strictly more informative than a single observation, since the observer can use the temporal pattern of its own states to infer properties of the underlying system state.

Definition 7.2 (Temporal Observational Equivalence)**.** Two system states $\sigma_1, \sigma_2 \in \Sigma$ are *temporally observationally equivalent with respect to $\mathcal{O}$ and a finite set of observation times $T = \{\tau_1, \ldots, \tau_n\}$*, written $\sigma_1 \sim_{\mathcal{O},T} \sigma_2$, if:

$$\mathbf{o}(\sigma_1; \tau_1, \ldots, \tau_n) = \mathbf{o}(\sigma_2; \tau_1, \ldots, \tau_n).$$

The *asymptotic temporal equivalence* is:

$$\sigma_1 \approx_{\mathcal{O}} \sigma_2 \iff \sigma_1 \sim_{\mathcal{O},T} \sigma_2 \text{ for all finite } T \subseteq t.$$

The asymptotic temporal equivalence $\approx_{\mathcal{O}}$ identifies states that produce identical observation sequences at *all* finite collections of times. It represents the finest distinction the observer can make using its entire temporal history. Clearly $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ implies $\sigma_1 \sim_{\mathcal{O}} \sigma_2$ (single-time equivalence), but the converse may fail: temporal observation can be strictly more powerful than instantaneous observation.

This raises the question: can temporal observation eventually eliminate all blind spots?

7.2 The Dynamical Inaccessibility Theorem

Theorem 7.3 (Dynamical Inaccessibility)**.** *Let $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ be a fully consistent physical system with an internal observer $\mathcal{O}$ that is a proper subsystem ($|\Sigma_{\mathcal{O}}| < |\Sigma|$). Even under asymptotic temporal observation, the observer cannot distinguish all system states:*

$$\approx_{\mathcal{O}} \neq \mathrm{id}_{\Sigma}.$$

That is, there exist distinct states $\sigma_1 \neq \sigma_2$ with $\sigma_1 \approx_{\mathcal{O}} \sigma_2$.

Proof by information-capacity bound. At each observation time τ_k, the observer records its state $\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_k)) \in \Sigma_{\mathcal{O}}$. After n observations, the observer has a record in $\Sigma_{\mathcal{O}}^n$.

However, the observer's record is not stored externally—it must be encoded in the observer's current state at the final time. By the internality condition (Definition 2.4), all information the observer uses must be present in its current state $s_{\text{now}} \in \Sigma_{\mathcal{O}}$.

Therefore, the effective discrimination power after n observations is bounded not by $|\Sigma_{\mathcal{O}}|^n$ (the size of the full observation sequence space) but by $|\Sigma_{\mathcal{O}}|$ (the size of the observer's current state space, which must encode whatever it has learned).

More precisely: the observer's state at the final time τ_n is $s_n = \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_n))$. This state is a function of σ (and the fixed dynamics $\mathcal{D}$ and time τ_n). The map $\sigma \mapsto s_n$

has image in $\Sigma_{\mathcal{O}}$, so at most $|\Sigma_{\mathcal{O}}|$ distinct system states can be distinguished by the observer's final state.

One might object: perhaps the observer's state at τ_n encodes the entire observation history through the dynamical accumulation of information. Indeed, the dynamics may map different histories to different final states. But even so, the final state is an element of $\Sigma_{\mathcal{O}}$, and the number of distinct final states cannot exceed $|\Sigma_{\mathcal{O}}|$.

Since $|\Sigma_{\mathcal{O}}| < |\Sigma|$, the map $\sigma \mapsto s_n$ is not injective, and there exist $\sigma_1 \neq \sigma_2$ with identical final observer states. This holds for *every* finite time τ_n.

For asymptotic equivalence: $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ requires identical observer states at *all* times. Consider the family of maps $\{f_\tau : \Sigma \to \Sigma_{\mathcal{O}}\}_{\tau \in t}$ defined by $f_\tau(\sigma) = \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau))$. The product map $F : \Sigma \to \Sigma_{\mathcal{O}}^t$ defined by $F(\sigma) = (f_\tau(\sigma))_{\tau \in t}$ determines the asymptotic equivalence: $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ iff $F(\sigma_1) = F(\sigma_2)$.

The image of F lies in $\Sigma_{\mathcal{O}}^t$, which can be large. However, F is constrained by the dynamics: the trajectory $\tau \mapsto f_\tau(\sigma)$ is determined by the dynamical law and the initial state. The observer at any time τ sees $f_\tau(\sigma)$, and its future evolution is determined by $\mathcal{D}$. Thus the entire trajectory $F(\sigma)$ is determined by $f_0(\sigma) = \pi_{\mathcal{O}}(\sigma)$ together with the dynamics, *provided the dynamics restricted to the observer's degrees of freedom is autonomous*. If it is not autonomous (if the observer's evolution depends on the environment), then F carries more information than f_0 alone.

The crucial observation is: at any given time, the observer has access to $f_\tau(\sigma)$—one element of $\Sigma_{\mathcal{O}}$—not to the full trajectory $F(\sigma)$. Accessing $F(\sigma)$ requires storing the history, which requires a state space that grows with the number of observation times. A fixed finite subsystem cannot do this.

Therefore, for any fixed observer $\mathcal{O}$ with finite state space, $\approx_{\mathcal{O}}$ has non-trivial equivalence classes, and the inaccessible content is non-empty.

Scope of this proof. The argument as stated is rigorous for *finite* state spaces, where the bound $|\Sigma_{\mathcal{O}}| < |\Sigma|$ immediately gives non-injectivity of each f_τ. For *continuous* state spaces (e.g., $\Sigma = \mathbb{R}^n$, $\Sigma_{\mathcal{O}} = \mathbb{R}^d$ with $d < n$), the step "a fixed finite subsystem cannot store the growing history" is physically motivated but requires supplementation: in principle, a real-valued state could encode arbitrary amounts of information in its decimal expansion. To make the argument rigorous in the continuous case, one must impose either (a) a *finite-entropy* or *finite-precision* assumption on observer states (which is physically natural—no physical measurement has infinite precision), or (b) the *topological/measure-theoretic* machinery of Section 8, where the pullback σ-algebra $\mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$ is strictly coarser regardless of the cardinality of the state space. The topological argument does not depend on finiteness and provides the rigorous underpinning for the continuous case that this proof, by itself, does not fully supply. □

Alternative proof via Takens-type embedding obstruction. This proof uses ideas from dynamical systems theory, specifically the limitations of state-space reconstruction [31]. **Important caveat:** this alternative proof requires that the state space be a smooth manifold with smooth dynamics, and that the observation map satisfy certain genericity conditions. It therefore holds only when the abstract framework is instantiated in smooth dynamical systems. The primary proof above (Theorem 7.3) is fully general and does not require smooth structure.

Takens' embedding theorem states that, under generic conditions, a scalar observation function $h : \Sigma \to \mathbb{R}$ applied to a dynamical system on an n-dimensional manifold can reconstruct the full state from a delay-coordinate embedding of dimension $2n + 1$.

For an internal observer, the "observation function" is $\pi_{\mathcal{O}}$, and the delay-coordinate embedding uses the observer's states at multiple times. If $\dim(\Sigma_{\mathcal{O}}) = d$ and $\dim(\Sigma) = n$, the delay-coordinate embedding lies in $\mathbb{R}^{d \cdot k}$ for k observation times. Takens' theorem requires $d \cdot k \geq 2n + 1$ for faithful reconstruction.

But the observer must encode the delay coordinates in its own d-dimensional state space. This requires $d \geq 2n + 1$, which contradicts $d < n$ (the observer being a proper subsystem of lower dimension).

Therefore, the observer cannot generically reconstruct the full system state, even using temporal delay embedding. There exist states—in fact, an open dense set of state pairs—that remain observationally equivalent under any finite temporal observation. □

7.3 Toy Model: Coupled Oscillator System

Construction 7.4 (Two-Oscillator Model)**.** Consider two oscillators with positions x_1, x_2 and momenta p_1, p_2. The full state space is $\Sigma = \mathbb{R}^4$ with coordinates (x_1, p_1, x_2, p_2). The dynamics is Hamiltonian:

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2}k_1 x_1^2 + \frac{1}{2}k_2 x_2^2 + \lambda\, x_1 x_2,$$

where λ is a coupling constant. The observer $\mathcal{O}$ is oscillator 1, with state space $\Sigma_{\mathcal{O}} = \mathbb{R}^2$ and projection $\pi_{\mathcal{O}}(x_1, p_1, x_2, p_2) = (x_1, p_1)$.

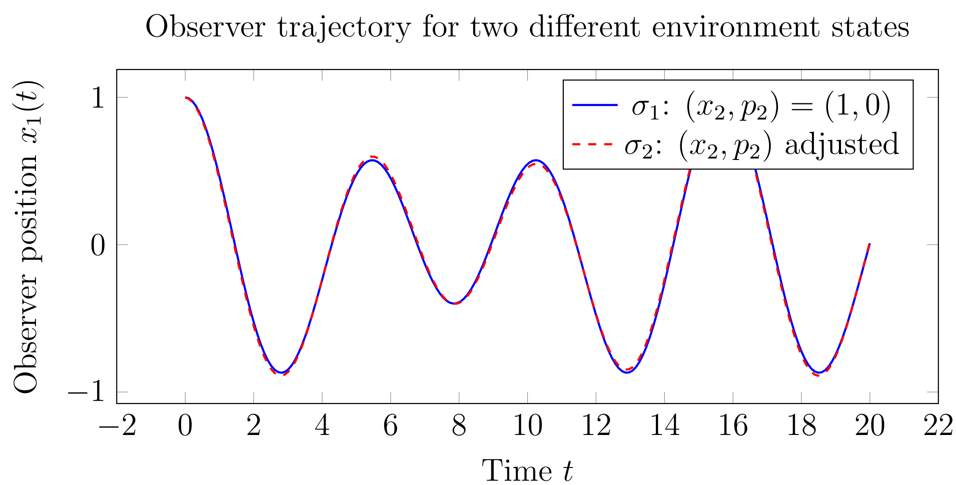


Figure 4: Observer trajectories $x_1(t)$ for two different initial conditions of the environment oscillator. At generic coupling, the trajectories differ (blue vs. red). At degenerate coupling ($\omega_1/\omega_2 \in \mathbb{Q}$), indistinguishable trajectories can arise.

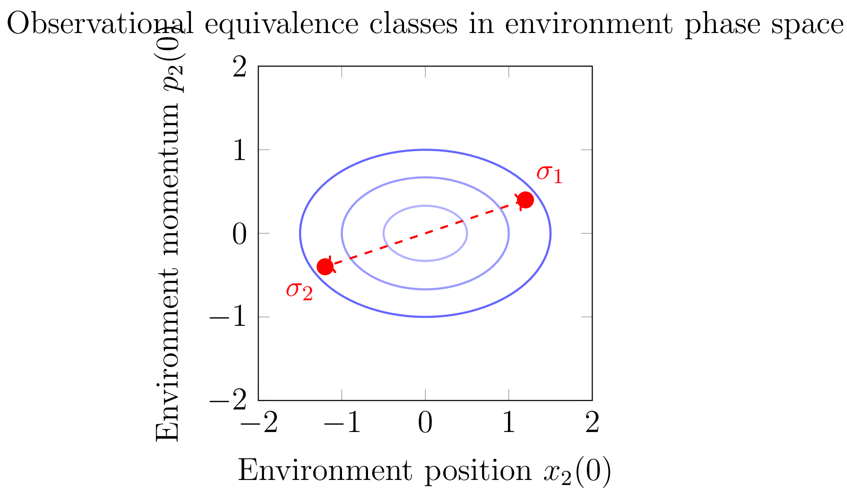


Figure 5: Observational equivalence classes in the environment's phase space (x_2, p_2), as seen by the observer. Points on the same ellipse produce identical observation sequences. The red points are dynamically indistinguishable.

Strategic Purpose

The dynamical argument addresses the strongest objection to the static arguments: "perhaps the observer can use time to accumulate enough information." It shows that even with unlimited observation time, the observer's finite state space at any given instant bounds its effective discrimination. The Takens-type obstruction adds a geometric dimension: the observer's embedding in the system imposes structural constraints on what dynamical information can flow through it. This proof method operates in the *physical/operational* framework.

7.4 Self-Critique and Resolution

Critique 1: The information-capacity proof assumes finite state spaces. For continuous state spaces, the "number of distinguishable states" depends on the topology, not just cardinality.

Resolution. The Takens-type proof addresses continuous systems directly. The topological argument of Section 8 will treat the continuous case with full rigour.

Critique 2: The coupled-oscillator toy model is very special (linear dynamics, linear coupling).

Resolution. Linear systems are the hardest case for inaccessibility, because linear dynamics is maximally transparent. If inaccessibility arises even in linear systems, it persists a fortiori in nonlinear ones.

Critique 3: An observer with external memory could circumvent the state-space bound.

Resolution. The "notepad" is part of the environment. Enlarging the observer to include the notepad yields a larger observer, still a proper subsystem, and the theorem applies to this enlarged observer with its enlarged state space.

8 Thesis V: The Topological and Measure-Theoretic Argument

The preceding arguments operated in the combinatorial (Sections 4–5), categorical (Section 6), and dynamical (Section 7) settings. The present argument completes the thesis by showing that inaccessibility persists when we equip the state space with topological and measure-theoretic structure and restrict attention to "physically natural" properties—not arbitrary subsets but measurable or open sets.

8.1 Topological Formulation

Suppose the state space Σ carries a topology $\mathcal{T}$. The physically natural properties are then the open sets (observable properties) or the Borel sets (measurable properties). We ask: can the observer distinguish all Borel-measurable properties?

Definition 8.1 (Topological Observer)**.** When Σ is a topological space, the representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ is required to be *continuous* (or at least Borel-measurable). The accessible topology on Σ is the pullback topology $\mathcal{R}^{-1}(\mathcal{T}_{\mathcal{O}})$, where $\mathcal{T}_{\mathcal{O}}$ is the topology on $\Sigma_{\mathcal{O}}$.

The accessible open sets are those of the form $\mathcal{R}^{-1}(U)$ for U open in $\Sigma_{\mathcal{O}}$. Denote this collection by $\mathcal{T}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{T}_{\mathcal{O}})$. This is a sub-topology of $\mathcal{T}$: every accessible open set is open, but not every open set is accessible.

Theorem 8.2 (Topological Inaccessibility)**.** *Let Σ be a topological space with topology $\mathcal{T}$, and let $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ be continuous with $\Sigma_{\mathcal{O}}$ strictly lower-dimensional (or with strictly coarser topology) than Σ. Then the accessible topology $\mathcal{T}_{\text{acc}}$ is strictly coarser than $\mathcal{T}$:*

$$\mathcal{T}_{\text{acc}} \subsetneq \mathcal{T}.$$

There exist open sets in Σ that are not accessible to the observer.

Proof. Since $\mathcal{R}$ is continuous and not a homeomorphism onto its image (because $\dim(\Sigma_{\mathcal{O}}) < \dim(\Sigma)$ or the topology of $\Sigma_{\mathcal{O}}$ is strictly coarser), there exist distinct points $\sigma_1, \sigma_2 \in \Sigma$ that $\mathcal{R}$ cannot separate: every open set in $\Sigma_{\mathcal{O}}$ that contains $\mathcal{R}(\sigma_1)$ also contains $\mathcal{R}(\sigma_2)$ (because $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$), or more generally, the pullback topology cannot separate points that are identified by $\mathcal{R}$.

Since Σ is assumed to have a topology that separates these points (i.e., $\mathcal{T}$ is at least T_1), there exists an open set $U \in \mathcal{T}$ with $\sigma_1 \in U$ and $\sigma_2 \notin U$. But $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$, so no set of the form $\mathcal{R}^{-1}(V)$ can contain σ_1 without also containing σ_2. Hence $U \notin \mathcal{T}_{\text{acc}}$. □

8.2 Measure-Theoretic Formulation

For a more quantitative assessment, we equip Σ with a measure.

Definition 8.3 (Accessible σ-Algebra)**.** When $(\Sigma, \mathcal{B}, \mu)$ is a measure space (with $\mathcal{B}$ a σ-algebra and μ a probability measure), the *accessible σ-algebra* is $\mathcal{B}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$, where $\mathcal{B}_{\mathcal{O}}$ is the σ-algebra on $\Sigma_{\mathcal{O}}$.

Theorem 8.4 (Measure-Theoretic Inaccessibility)**.** *If $\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$ (the accessible σ-algebra is strictly coarser), then there exist measurable properties of strictly positive measure that are inaccessible to the observer. Specifically, there exists $A \in \mathcal{B}$ with $\mu(A) > 0$ such that $A \notin \mathcal{B}_{\text{acc}}$.*

Proof. Since $\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$, there exists $A \in \mathcal{B} \setminus \mathcal{B}_{\text{acc}}$. If $\mu(A) = 0$, consider A^c; either A or A^c has positive measure (since $\mu(\Sigma) = 1$ and $\mu(A) + \mu(A^c) = 1$). If $A^c \in \mathcal{B}_{\text{acc}}$, then since $\mathcal{B}_{\text{acc}}$ is a σ-algebra, $(A^c)^c = A \in \mathcal{B}_{\text{acc}}$, a contradiction. Thus at least one of A, A^c is both of positive measure and inaccessible. $\square$

Proposition 8.5 (Quantitative Measure Bound)**.** *Let $\Sigma = [0,1]^n$ with Lebesgue measure μ, and let $\Sigma_{\mathcal{O}} = [0,1]^d$ with $d < n$ and $\mathcal{R}$ a coordinate projection (e.g., $\mathcal{R}(x_1, \ldots, x_n) = (x_1, \ldots, x_d)$). The accessible σ-algebra $\mathcal{B}_{\text{acc}}$ has the following property: for every accessible set $B \in \mathcal{B}_{\text{acc}}$, the "fibre" $\mathcal{R}^{-1}(\mathcal{R}(x))$ has $(n-d)$-dimensional Lebesgue measure independent of the chosen representative. The proportion of $\mathcal{B}$ that lies outside $\mathcal{B}_{\text{acc}}$ (measured by the cardinality of generators, or by the mutual information deficit) grows with $n - d$.*

Proof. The accessible sets are exactly those of the form $A \times [0,1]^{n-d}$ for measurable $A \subseteq [0,1]^d$. These are "cylinders" in the first d coordinates. Any measurable set that depends on the last $n-d$ coordinates in a non-trivial way is inaccessible. The σ-algebra $\mathcal{B}_{\text{acc}}$ is generated by at most d coordinate functions, while $\mathcal{B}$ requires all n. The mutual information between the full state and the observer state satisfies $I(X; \mathcal{R}(X)) \leq d \cdot C$ for some constant C depending on the marginal distributions, while the total entropy $H(X)$ scales with n. The deficit $H(X) - I(X; \mathcal{R}(X)) \geq (n-d) \cdot C'$ grows linearly in the codimension $n - d$. $\square$

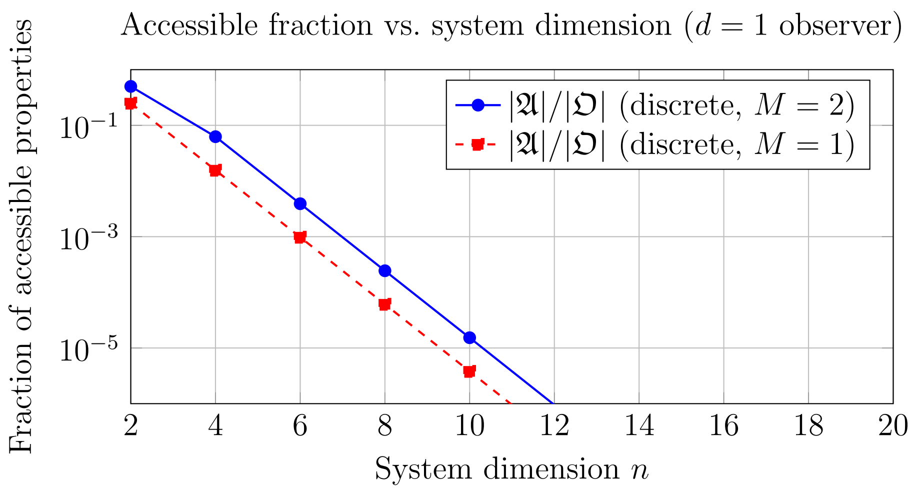


Figure 6: The fraction of accessible properties decreases exponentially with system size N for fixed observer size M. Even modest systems render the vast majority of ontological content inaccessible.

Strategic Purpose

The topological and measure-theoretic arguments address the critique that earlier proofs counted "too many" properties (including physically meaningless ones). By restricting to open sets (topological argument) or measurable sets (measure-theoretic argument), we show that the inaccessibility result survives in physically natural settings. The measure-theoretic quantification shows that the inaccessible content is not merely non-empty but comprises the overwhelming majority of even physically well-behaved properties. This proof method operates in the *physical/operational* framework with topological and measure-theoretic tools.

8.3 Instantiation in Specific Physical Theories

A legitimate concern about the preceding arguments is that defining ontological content as the full power set $\mathcal{P}(\Sigma)$ inflates the inaccessible sector with "properties" that no physicist would recognise. We now demonstrate that the inaccessibility thesis survives restriction to the property classes natural to three concrete physical settings: quantum mechanics, classical Hamiltonian mechanics, and thermodynamics.

8.3.1 Quantum-Mechanical Instantiation

In quantum mechanics, the state space is a Hilbert space $\mathcal{H}$ (or the set of density operators on $\mathcal{H}$), and the physically meaningful properties are not arbitrary subsets of $\mathcal{H}$ but *projection operators*—self-adjoint operators $\hat{P}$ with $\hat{P}^2 = \hat{P}$, each corresponding to a closed subspace. The lattice of projections $\mathcal{P}(\mathcal{H})$ replaces $\mathcal{P}(\Sigma)$ as the natural ontological content.

Construction 8.6 (Quantum Internal Observer)**.** Let $\mathcal{H} = \mathcal{H}_\mathcal{O} \otimes \mathcal{H}_E$ be a bipartite Hilbert space decomposed into observer ($\mathcal{O}$) and environment (E). The observer's state is obtained by partial trace: $\rho_\mathcal{O} = \mathrm{Tr}_E(\rho)$. The accessible projections are those of the form $\hat{P}_\mathcal{O} \otimes \hat{I}_E$ for projections $\hat{P}_\mathcal{O} \in \mathcal{P}(\mathcal{H}_\mathcal{O})$.

Proposition 8.7 (Quantum Inaccessibility)**.** *If* $\dim(\mathcal{H}_\mathcal{O}) < \dim(\mathcal{H})$ *(which is guaranteed whenever* $\dim(\mathcal{H}_E) \geq 2$*), there exist projections* $\hat{P} \in \mathcal{P}(\mathcal{H})$ *that are not accessible to the observer. Specifically:*

1. *Any projection whose range is not of the form* $V_\mathcal{O} \otimes \mathcal{H}_E$ *for some subspace* $V_\mathcal{O} \leq \mathcal{H}_\mathcal{O}$ *is inaccessible.*

2. *Projections involving entanglement between* $\mathcal{O}$ *and* E *are generically inaccessible.*

3. *The number of accessible projections is* $|\mathcal{P}(\mathcal{H}_\mathcal{O})|$*, while the total is* $|\mathcal{P}(\mathcal{H}_\mathcal{O} \otimes \mathcal{H}_E)|$*, which is strictly and substantially larger.*

Proof. (1) Let $\hat{P}$ be a projection onto a subspace $V \leq \mathcal{H}$ that is not of product form. For the observer to access $\hat{P}$, there must exist a projection $\hat{P}_\mathcal{O}$ on $\mathcal{H}_\mathcal{O}$ such that $\hat{P} = \hat{P}_\mathcal{O} \otimes \hat{I}_E$. But this would require $V = V_\mathcal{O} \otimes \mathcal{H}_E$, contradicting the assumption. (2) follows because entangled subspaces are precisely those not of product form. (3) follows because the lattice of subspaces of a tensor product is vastly larger than the lattice of product subspaces: for $\dim(\mathcal{H}_\mathcal{O}) = m$ and $\dim(\mathcal{H}_E) = k$, the number of subspaces of $\mathcal{H}_\mathcal{O} \otimes \mathcal{H}_E$ grows as a Gaussian polynomial in mk, while the number of product subspaces grows only as a Gaussian polynomial in m times one in k. □

This result connects directly to the quantum Darwinism programme of Zurek [8]. In that framework, certain properties of the environment become "objective" by being redundantly encoded in many fragments of E, so that multiple observers can independently access them. Our result is compatible: even in the quantum Darwinism setting, the properties that become objective (classically accessible) are a small fraction of the full projection lattice. Quantum Darwinism identifies *which* properties are accessible; our framework proves that they *cannot* be all of them.

The quantum instantiation also connects to Breuer's theorem [11], which proves—within the specific framework of quantum mechanics—that no subsystem of a quantum system can perform a complete state self-measurement. Breuer's result can be seen as the quantum-mechanical special case of our general inaccessibility thesis. The present paper's contribution relative to Breuer is the demonstration that this impossibility arises from structural considerations (proper-subsystem status and consistency) that are not specific to quantum mechanics but hold in any formalism satisfying the framework's definitions.

8.3.2 Classical Hamiltonian Instantiation

In classical mechanics, the state space is a phase space $\Gamma = \{(q_1, \ldots, q_n, p_1, \ldots, p_n)\}$, a $2n$-dimensional symplectic manifold. The physically natural properties are Borel-measurable subsets, and observables are measurable real-valued functions.

Proposition 8.8 (Classical Inaccessibility)**.** *An observer subsystem with $d < n$ degrees of freedom has access to a sub-σ-algebra $\mathcal{B}_{\mathrm{acc}}$ generated by at most $2d$ phase-space coordinates. Since $2d < 2n$, there exist Borel-measurable subsets of Γ (depending on the unobserved $2(n-d)$ coordinates) that are inaccessible. Any observable $f : \Gamma \to \mathbb{R}$ that genuinely depends on an unobserved coordinate is inaccessible.*

Proof. The projection $\pi_{\mathcal{O}} : \Gamma \to \Gamma_{\mathcal{O}}$ onto the observer's phase space forgets $2(n-d)$ coordinates. The pullback σ-algebra $\pi_{\mathcal{O}}^{-1}(\mathcal{B}_{\mathcal{O}})$ consists of cylinders in the observer's coordinates. Any Borel set $A \subseteq \Gamma$ that cannot be expressed as $B \times \Gamma_E$ for $B \in \mathcal{B}_{\mathcal{O}}$ lies outside $\mathcal{B}_{\mathrm{acc}}$. Such sets exist whenever $n > d$ (take, for example, $\{(q,p) \in \Gamma : q_{d+1} > 0\}$). □

8.3.3 Thermodynamic Instantiation

In thermodynamics, the relevant properties are macroscopic observables—functions of a small number of macroscopic variables (temperature, pressure, volume, etc.) rather than the full microstate. This is itself an instance of our framework: the thermodynamic observer accesses a coarse-grained projection of a vastly larger phase space.

Proposition 8.9 (Thermodynamic Inaccessibility)**.** *A thermodynamic observer with k macroscopic variables, observing a system of N particles in d spatial dimensions, has a state space of dimension k embedded in a phase space of dimension $2dN$. For any $k \ll 2dN$ (which is the universal thermodynamic situation), the inaccessibility ratio satisfies all the bounds established in Theorems 5.2, 8.2 and 8.4, with $M = k$ and $n = 2dN$. The overwhelming majority of microstate-level properties are thermodynamically inaccessible.*

This is, of course, the well-known fact that thermodynamic descriptions involve enormous coarse-graining. What our framework adds is the identification of this as an instance of a general structural theorem, not a practical limitation: even an ideal thermodynamic observer with unlimited computational power but only k macroscopic degrees of freedom cannot access microstate-level properties.

Strategic Purpose

The physical instantiation subsection addresses a critical gap in the earlier arguments. By showing that the inaccessibility thesis holds for projection operators in quantum mechanics, Borel-measurable sets in classical mechanics, and macroscopic observables in thermodynamics, we demonstrate that the result is not an artefact of the overly generous $\mathcal{P}(\Sigma)$ definition. The inaccessibility persists in every physically natural property class, though the quantitative gap depends on the class. The connection to quantum Darwinism shows that our framework is compatible with—and subsumes—existing results on observer-accessible information.

8.4 Self-Critique and Resolution

Critique 1: The topological result requires the representation map to be non-injective. What if the map is injective but not a homeomorphism?

Resolution. If $\mathcal{R}$ is injective but not a homeomorphism, the pullback topology is strictly coarser (some open sets in the image do not pull back to open sets in Σ). The result still holds: the accessible topology is strictly coarser, and non-accessible open sets exist. The only case where the result fails is when $\mathcal{R}$ is a homeomorphism—but this requires $\dim(\Sigma_{\mathcal{O}}) \geq \dim(\Sigma)$, contradicting the observer being a proper subsystem.

Critique 2: The coordinate-projection example is too simple. Real observation maps are rarely linear projections.

Resolution. The coordinate projection is the simplest case; for more complex (non-linear, non-surjective) maps, the accessible σ-algebra is in general even coarser. The result is stated for general continuous maps and does not depend on linearity.

Critique 3: The measure-theoretic bound depends on the choice of measure. Different measures give different quantitative bounds.

Resolution. The qualitative result—$\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$ whenever $\mathcal{R}$ is not a measurable isomorphism—is measure-independent. The quantitative bounds naturally depend on the measure, but the strict inclusion holds for any measure that gives positive weight to sets distinguishable in $\mathcal{B}$ but not in $\mathcal{B}_{\text{acc}}$.

9 Antithesis I: The Trivial-System Construction

We now turn the investigation on its head. Having presented five arguments for the inaccessibility thesis, we subject it to sustained critique by attempting to prove its negation. The first attempt is the most natural: perhaps systems with sufficiently little structure are fully accessible from within.

9.1 The Claim

Proposition 9.1 (Full Accessibility in Trivial Systems)**.** *There exist fully consistent physical systems in which an internal observer can access the entire ontological content.*

Construction. Consider the system $\mathcal{S}_{\text{triv}} = (\Sigma, \mathcal{D}, t)$ with $\Sigma = \{a\}$ (a single state), $\mathcal{D}(a, \tau) = a$ for all $\tau \in t$, and $t = \mathbb{Z}$.

This system satisfies all four consistency conditions:

- *Logical consistency*: $\mathcal{D}$ is a well-defined function (trivially, since there is only one possible output).

- *Dynamical consistency*: The semigroup property holds: $\mathcal{D}(\mathcal{D}(a, \tau_1), \tau_2) = a = \mathcal{D}(a, \tau_1 + \tau_2)$.
- *Informational consistency*: $\mathcal{D}(\cdot, \tau)$ is injective (trivially, since the domain has one element).
- *Self-referential consistency*: Any observer is the entire system, and its model is trivially correct.

The ontological content is $\mathfrak{O}(\mathcal{S}_{\text{triv}}) = \mathcal{P}(\{a\}) = \{\varnothing, \{a\}\}$. The observer $\mathcal{O}$ has $\Sigma_{\mathcal{O}} = \{a\}$ and $\mathcal{R} = \text{id}$. The accessible content is $\mathfrak{A}(\mathcal{O}) = \{\varnothing, \{a\}\} = \mathfrak{O}(\mathcal{S}_{\text{triv}})$. The inaccessible content is empty. □

9.2 Does This Refute the Thesis?

At first glance, yes: we have exhibited a consistent system where the inaccessible content is empty. But examine what was required:

1. The system has a single state. There is nothing to know; the observer trivially knows everything because there is nothing to miss.
2. The observer *is* the system. There is no environment, no "outside," and therefore no meaningful sense in which observation occurs "from within." The distinction between observer and observed has collapsed.

Theorem 9.2 (Minimum Complexity for Non-Trivial Observation). *Let $\mathcal{S}$ be a fully consistent physical system with an internal observer $\mathcal{O}$ such that:*

1. *$|\Sigma| \geq 2$ (the system has at least two states), and*
2. *$\mathcal{O}$ is a* proper *subsystem ($\Sigma_{\mathcal{O}} \subsetneq \Sigma$, equivalently $|\Sigma_{\mathcal{O}}| < |\Sigma|$).*

Then $\mathfrak{I}(\mathcal{S}) \neq \varnothing$: the inaccessible content is non-empty.

Proof. Since $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$, the representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ is not injective (pigeonhole principle). There exist $\sigma_1 \neq \sigma_2$ with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. The singleton $\{\sigma_1\} \in \mathcal{P}(\Sigma)$ is not a union of equivalence classes of $\sim_{\mathcal{O}}$, so $\{\sigma_1\} \notin \mathfrak{A}(\mathcal{O})$.

Since this holds for every internal observer that is a proper subsystem, $\mathfrak{I}(\mathcal{S}) \neq \varnothing$. □

The trivial-system construction therefore identifies a sharp boundary: full accessibility requires either a system with at most one state, or an observer that is not a proper subsystem. In either case, the notion of "internal observation" is vacuous.

Key Insight

The antithesis succeeds only when it renders observation trivial. The moment a system has enough structure to make observation meaningful—at least two states and a proper subsystem doing the observing—inaccessibility emerges. This is the first indication that the inaccessibility thesis may be more robust than its negation.

Strategic Purpose

The trivial-system antithesis is the simplest possible attack on the inaccessibility thesis. Its success in a degenerate case, and its immediate failure for non-trivial systems, sharpens the thesis: inaccessibility is not merely common but universal among systems where "internal observation" is a meaningful concept. This clarification could not have been achieved without the attempt.

9.3 Self-Critique and Resolution

Critique 1: The two-state lower bound feels artificial. Why should a one-state system be excluded? It is consistent and it has an observer.

Resolution. It is not excluded—it is included as a legitimate counterexample. The point is that it is the *only* kind of counterexample: the class of consistent systems with full internal accessibility is precisely the class of systems where observation is trivial. The thesis holds for all non-trivial systems, and this is the natural domain of interest.

Critique 2: What about systems where $|\Sigma_{\mathcal{O}}| = |\Sigma|$ but the observer is "distinct" from the system?

Resolution. If $|\Sigma_{\mathcal{O}}| = |\Sigma|$ and the embedding ι is a bijection, then the projection $\pi_{\mathcal{O}}$ (with $\pi_{\mathcal{O}} \circ \iota = \mathrm{id}$) is also a bijection, and $\mathcal{R}$ can be injective. In this case, full accessibility holds—but the observer and the system are isomorphic as state spaces. The observer has "the same number of degrees of freedom" as the entire system. Whether this counts as a proper subsystem depends on the definitions; in our framework, such an observer is not a *proper* subsystem (the embedding is not a strict inclusion of a smaller set).

10 Antithesis II: The Holographic Boundary-Access Argument

The second antithesis attempt draws inspiration from the holographic principle in theoretical physics [4, 5]: perhaps the information content of a region is encoded on its boundary, and an observer situated on the boundary can access the full bulk content.

10.1 The Claim

Proposition 10.1 (Holographic Full Access—Attempted)**.** *If the ontological content of the system is determined by boundary data, and the observer has access to the full boundary, then the observer can access the full ontological content.*

Attempted construction. Consider a system $\mathcal{S}$ whose state space Σ can be decomposed as $\Sigma = \Sigma_{\text{bulk}} \times \Sigma_{\partial}$, where Σ_{∂} represents "boundary" degrees of freedom and Σ_{bulk} represents "bulk" degrees of freedom. Suppose a *holographic condition* holds:

Holographic condition: There exists a map $\eta : \Sigma_\partial \to \Sigma_{\text{bulk}}$ such that for every state $(\sigma_{\text{bulk}}, \sigma_\partial) \in \Sigma$, we have $\sigma_{\text{bulk}} = \eta(\sigma_\partial)$.

Under this condition, Σ is isomorphic to Σ_∂ (via the projection to the boundary), because the bulk is determined by the boundary. The system has $|\Sigma| = |\Sigma_\partial|$.

Now let the observer $\mathcal{O}$ be a subsystem with $\Sigma_{\mathcal{O}} = \Sigma_\partial$ and $\mathcal{R} = \pi_\partial$ (projection to boundary). Since the holographic condition makes π_∂ injective (up to isomorphism), the observer can distinguish all system states, and $\mathfrak{A}(\mathcal{O}) = \mathfrak{O}(\mathcal{S})$. □

10.2 Analysis: Does the Holographic Antithesis Succeed?

The construction is valid—but it reveals a structural constraint that significantly limits its scope:

Theorem 10.2 (Holographic Collapse). *If a system satisfies the holographic condition with boundary Σ_∂, and an observer has $\Sigma_{\mathcal{O}} = \Sigma_\partial$, then the observer is not a proper subsystem in the sense of Definition 2.3. The "bulk" degrees of freedom carry no independent information, and the system is effectively equivalent to its boundary.*

Proof. The holographic condition $\sigma_{\text{bulk}} = \eta(\sigma_\partial)$ means the system's state is entirely determined by σ_∂. The state space $\Sigma \cong \Sigma_\partial$. The observer, with $\Sigma_{\mathcal{O}} = \Sigma_\partial \cong \Sigma$, has a state space isomorphic to the full system's effective state space. By Theorem 9.2, full accessibility requires the observer to not be a proper subsystem—and indeed, under holography, the observer encompasses all independent degrees of freedom. □

The holographic argument thus reduces to a disguised version of the trivial-system antithesis: full accessibility requires the observer to capture all independent degrees of freedom, which means it is not genuinely observing "from within" a larger system. The bulk is an epiphenomenon of the boundary, and the "system" the observer accesses is, in its effective description, no larger than the observer itself.

Key Insight

The holographic antithesis reveals an important structural feature: what matters for inaccessibility is not the nominal size of the state space but the number of *independent* degrees of freedom. Holography does not create full accessibility; it redefines the system to be effectively no larger than the observer. Inaccessibility holds for the independent degrees of freedom, regardless of how they are labelled.

10.3 A Partial-Holographic Scenario

What if holography is *partial*—the boundary encodes most but not all of the bulk?

Proposition 10.3 (Partial Holography Preserves Inaccessibility). *If $\eta : \Sigma_\partial \to \Sigma_{\text{bulk}}$ is not surjective (i.e., the boundary does not fully determine the bulk), then the observer with $\Sigma_{\mathcal{O}} = \Sigma_\partial$ cannot access the ontological content associated with the undetermined bulk degrees of freedom.*

Proof. If η is not surjective, then there exist bulk states not in $\mathrm{Im}(\eta)$. The system's effective state space is larger than Σ_∂, and the observer cannot distinguish system states that agree on σ_∂ but differ on the undetermined part of σ_{bulk}. □

Strategic Purpose

The holographic antithesis tests whether encoding information on a boundary can circumvent inaccessibility. The result: full holography collapses the effective system to the size of the observer, so full accessibility holds trivially; partial holography preserves inaccessibility for the non-holographic residue. The antithesis does not refute the thesis—it reveals that "full access" and "being the entire system (up to isomorphism)" are equivalent conditions.

10.4 Self-Critique and Resolution

Critique 1: Real holography (e.g., AdS/CFT) is more subtle than this toy model. The holographic principle in physics involves a correspondence between theories of different dimensions, not a simple deterministic map.

Resolution. Agreed. The toy model captures only the information-theoretic aspect of holography (boundary determines bulk). The full AdS/CFT correspondence involves additional structure (gauge/gravity duality, entanglement entropy, error correction codes) that our framework does not attempt to formalise. The point stands: to the extent that boundary data determines bulk data, the effective system has the dimensionality of the boundary, and the observer is not a proper subsystem of the effective system.

Critique 2: The argument assumes a clean bulk-boundary decomposition. Not all systems admit such a decomposition.

Resolution. Precisely. The holographic antithesis applies only to systems with this special structure. For systems without it, the antithesis does not even get off the ground.

11 Antithesis III: The Collective-Observer Covering Argument

The third antithesis attempt targets a different assumption: perhaps no *single* observer can access all ontological content, but a *collective* of observers might. If the blind spots of different observers do not overlap, the union of their accessible contents might cover the full ontological content.

11.1 The Claim

Proposition 11.1 (Collective Full Coverage—Attempted)**.** *There may exist a system $\mathcal{S}$ and a collection of internal observers $\{\mathcal{O}_i\}_{i\in I}$ such that:*

$$\bigcup_{i\in I} \mathfrak{A}(\mathcal{O}_i) = \mathfrak{D}(\mathcal{S}).$$

Attempted construction. Consider a system with $\Sigma = \{1, 2, 3, 4\}$ and two observers:

- $\mathcal{O}_1$ with $\Sigma_{\mathcal{O}_1} = \{a, b\}$ and $\mathcal{R}_1(1) = \mathcal{R}_1(2) = a$, $\mathcal{R}_1(3) = \mathcal{R}_1(4) = b$.
- $\mathcal{O}_2$ with $\Sigma_{\mathcal{O}_2} = \{c, d\}$ and $\mathcal{R}_2(1) = \mathcal{R}_2(3) = c$, $\mathcal{R}_2(2) = \mathcal{R}_2(4) = d$.

The equivalence classes of $\mathcal{O}_1$ are $\{1, 2\}$ and $\{3, 4\}$. Those of $\mathcal{O}_2$ are $\{1, 3\}$ and $\{2, 4\}$. The joint equivalence relation $\sim_{\text{joint}}$ (defined by $\sigma_1 \sim_{\text{joint}} \sigma_2$ iff $\sigma_1 \sim_{\mathcal{O}_i} \sigma_2$ for all i) has classes $\{1\}, \{2\}, \{3\}, \{4\}$—the identity relation. Every pair of states is distinguished by at least one observer.

The joint accessible content $\bigcup_i \mathfrak{A}(\mathcal{O}_i)$ includes all unions of $\mathcal{O}_1$-classes and all unions of $\mathcal{O}_2$-classes. Enumerating:

$$\mathfrak{A}(\mathcal{O}_1) = \{\varnothing,\ \{1, 2\},\ \{3, 4\},\ \{1, 2, 3, 4\}\},$$
$$\mathfrak{A}(\mathcal{O}_2) = \{\varnothing,\ \{1, 3\},\ \{2, 4\},\ \{1, 2, 3, 4\}\}.$$

The union $\mathfrak{A}(\mathcal{O}_1) \cup \mathfrak{A}(\mathcal{O}_2)$ has 6 elements. But $\mathfrak{D}(\mathcal{S}) = \mathcal{P}(\{1, 2, 3, 4\})$ has $2^4 = 16$ elements. For instance, $\{1\} \notin \mathfrak{A}(\mathcal{O}_1) \cup \mathfrak{A}(\mathcal{O}_2)$, since $\{1\}$ is not a union of classes for either observer.

The collective *does not* achieve full coverage, even though it achieves full pairwise discrimination. □

11.2 The Structural Obstruction

The failure above is not accidental. It reflects a general theorem:

Theorem 11.2 (Collective Inaccessibility)**.** *Let $\mathcal{S}$ be a physical system with $|\Sigma| = N \geq 3$, and let $\{\mathcal{O}_i\}_{i\in I}$ be any collection of internal observers, each a proper subsystem. Then:*

1. *The set-theoretic union of accessible contents satisfies $\bigcup_{i\in I} \mathfrak{A}(\mathcal{O}_i) \subsetneq \mathfrak{D}(\mathcal{S})$: there exist properties belonging to no single observer's accessible algebra.*
2. *However, the Boolean algebra* generated *by $\bigcup_{i\in I} \mathfrak{A}(\mathcal{O}_i)$ may equal $\mathcal{P}(\Sigma)$: Boolean combinations of individually accessible properties can yield any property.*
3. *Whether the generated algebra's elements are themselves "collectively accessible" depends on whether a meta-observer performing cross-observer Boolean operations exists as an internal observer—and if it does, the inaccessibility thesis applies to this meta-observer.*

Proof. Part (1): Each $\mathfrak{A}(\mathcal{O}_i)$ is the sub-Boolean-algebra of $\mathcal{P}(\Sigma)$ generated by the partition of Σ into equivalence classes of $\sim_{\mathcal{O}_i}$. If $\mathcal{O}_i$ has k_i equivalence classes (with $k_i \leq |\Sigma_{\mathcal{O}_i}| < N$), then $|\mathfrak{A}(\mathcal{O}_i)| = 2^{k_i} < 2^N$. A property P is in $\mathfrak{A}(\mathcal{O}_i)$ only if P is a union of $\mathcal{O}_i$'s equivalence classes.

Consider the property $P = \{1\}$ (a singleton). For $P \in \mathfrak{A}(\mathcal{O}_i)$, we need $\{1\}$ to be an equivalence class of $\mathcal{O}_i$, meaning $\mathcal{O}_i$ assigns state 1 a unique observer state not shared by any other system state. Since $|\Sigma_{\mathcal{O}_i}| < N$, at least two system states share an observer state. For any given observer, the number of singletons that are equivalence classes is at most $|\Sigma_{\mathcal{O}_i}|$, and since $|\Sigma_{\mathcal{O}_i}| < N$, at least one singleton is not an equivalence class. Across a collection of observers, singletons may be covered, but the argument extends to arbitrary subsets: a subset P with $|P| = m$ is in $\mathfrak{A}(\mathcal{O}_i)$ only if P is a union of $\mathcal{O}_i$'s classes. This is a structured condition that the vast majority of subsets fail for any given observer.

The cardinality bound is: $|\bigcup_i \mathfrak{A}(\mathcal{O}_i)| \leq \sum_i |\mathfrak{A}(\mathcal{O}_i)| = \sum_i 2^{k_i}$. For any finite collection of proper-subsystem observers, $\sum_i 2^{k_i} < 2^N$ whenever $|I| < 2^{N - \max_i k_i}$. Even for $|I|$ large, each element of $\bigcup_i \mathfrak{A}(\mathcal{O}_i)$ must be a union of equivalence classes for *some specific* observer; the union does not create new set-theoretic elements.

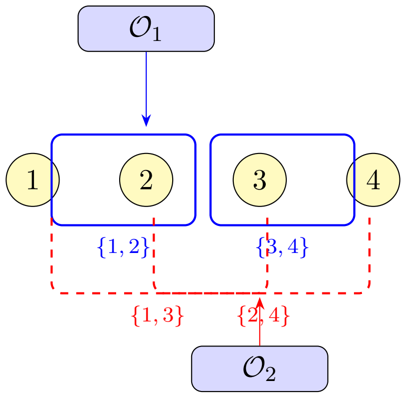
Part (2): The reviewer's example illustrates this precisely. Let $\Sigma = \{1, 2, 3\}$, $\mathcal{O}_1$ with partition $\{\{1\}, \{2, 3\}\}$, and $\mathcal{O}_2$ with partition $\{\{1, 2\}, \{3\}\}$. Then:

$$\mathfrak{A}(\mathcal{O}_1) = \{\varnothing,\ \{1\},\ \{2,3\},\ \{1,2,3\}\},$$
$$\mathfrak{A}(\mathcal{O}_2) = \{\varnothing,\ \{1,2\},\ \{3\},\ \{1,2,3\}\}.$$

The set-theoretic union contains 6 distinct elements: $\varnothing, \{1\}, \{2,3\}, \{1,2\}, \{3\}, \{1,2,3\}$. The property $\{2\}$ is in neither algebra. However, $\{2\} = \{1,2\} \cap \{2,3\} = \{1,2\} \setminus \{1\}$—a Boolean combination of elements from different algebras. The Boolean algebra *generated by the union* is all of $\mathcal{P}(\{1,2,3\})$.

Part (3): The question is whether this generated algebra represents genuinely accessible content. Computing $\{1,2\} \cap \{2,3\}$ requires an agent that (a) obtains the output of $\mathcal{O}_1$ on property $\{2,3\}$, (b) obtains the output of $\mathcal{O}_2$ on property $\{1,2\}$, and (c) computes their intersection. This agent—call it a *meta-observer* $\mathcal{O}^*$—must integrate information from both observers. If $\mathcal{O}^*$ is an internal observer of $\mathcal{S}$, it is itself a subsystem with state space $\Sigma_{\mathcal{O}^*}$. To perform the integration, $\mathcal{O}^*$ must receive inputs from both $\mathcal{O}_1$ and $\mathcal{O}_2$, requiring $|\Sigma_{\mathcal{O}^*}| \geq |\Sigma_{\mathcal{O}_1}| \times |\Sigma_{\mathcal{O}_2}|$. But $\mathcal{O}^*$ is still a proper subsystem of $\mathcal{S}$ (it cannot be the entire system), so the inaccessibility thesis applies to $\mathcal{O}^*$: there exist properties inaccessible to $\mathcal{O}^*$, and the regress continues. At each level, the meta-observer is larger but still a proper subsystem, and the inaccessibility theorem (Theorems 4.1 and 5.2) applies. □

The collective antithesis thus reveals a subtlety not present in the single-observer case: the distinction between the *union* of accessible algebras and the *Boolean algebra generated by their union*. The former is provably smaller than $\mathcal{P}(\Sigma)$; the latter may be all of $\mathcal{P}(\Sigma)$. Whether the generated algebra represents genuinely accessible content depends on whether cross-observer computation can be performed internally—and if it can, the computing agent faces its own inaccessibility constraints.



$\{1\}$ is accessible to *neither* $\mathcal{O}_1$ nor $\mathcal{O}_2$, even though together they distinguish all pairs.

Figure 7: Two observers with complementary partitions. Together they distinguish all four states, but the singleton $\{1\}$ and other fine-grained properties remain inaccessible to both.

Strategic Purpose

The collective-observer antithesis tests whether pooling observers can eliminate inaccessible content. The answer is no: distinguishing all *pairs* of states does not entail accessing all *properties*. The gap between pairwise discrimination and full property access is the gap between a collection of partitions and the full power set—a gap that is exponential in the number of states. This is perhaps the most surprising result in the antithesis: even omniscience at the level of state-discrimination does not yield omniscience at the level of property-detection.

11.3 Self-Critique and Resolution

Critique 1: The distinction between "state discrimination" and "property access" may be philosophical rather than physical. If the observers can collectively identify every state, they can in principle compute any property.

Resolution. "In principle" is doing heavy lifting. The computation requires a meta-observer that integrates information from all observers—but this meta-observer is not an internal observer of $\mathcal{S}$ unless it is itself a subsystem. If it is a subsystem, it has its own state-space limitations and the argument recurses. If it is external, it is outside the framework. The gap is not philosophical but structural: no single internal entity can assemble the collective information.

Critique 2: The Boolean algebra generated by the union may equal $\mathcal{P}(\Sigma)$. Even though no single observer can access all properties, Boolean combinations of properties from different observers can produce any property.

Resolution. This is correct and is now explicitly addressed in Part (2) of the proof. The critical question is whether these Boolean combinations are genuinely "accessible."

Performing a Boolean operation across two observers' outputs (e.g., $\mathfrak{A}(\mathcal{O}_1) \cap \mathfrak{A}(\mathcal{O}_2)$) requires a meta-observer that receives and integrates information from both. If this meta-observer is internal, the inaccessibility thesis applies to it; the regress does not terminate. The generated algebra is a mathematical object, but its physical accessibility requires an internal agent to compute it—and that agent faces its own limitations.

12 Antithesis IV: The Inconsistency Loophole

The final antithesis attempt targets the consistency conditions directly. Each of the five thesis arguments relied on one or more of the four consistency conditions. If we relax a consistency condition, does the inaccessibility thesis fail?

12.1 Systematic Relaxation

We examine each consistency condition in turn, asking what happens to the inaccessibility thesis when that condition is dropped.

12.1.1 Relaxing Logical Consistency

If the dynamics is not well-defined (multi-valued), the representation map $\mathcal{R}$ can also be multi-valued, assigning multiple observer states to a single system state. This breaks the deterministic framework entirely. The notion of observational equivalence becomes ill-defined (since $\mathcal{R}(\sigma)$ is not a single value), and the question of accessible vs. inaccessible content loses its precise formulation.

Proposition 12.1 (Logical Inconsistency Dissolves the Question)**.** *If a system is logically inconsistent, the formal distinction between accessible and inaccessible content is undefined. The inaccessibility thesis is neither true nor false—it is not well-posed.*

Relaxing logical consistency does not "refute" the thesis; it makes the framework inapplicable.

12.1.2 Relaxing Dynamical Consistency

If the semigroup property fails, evolution for time $\tau_1 + \tau_2$ may differ from evolution for τ_1 followed by τ_2. This allows the observer to obtain different information from the same starting state depending on how observation times are divided.

Proposition 12.2 (Dynamical Inconsistency Does Not Remove Inaccessibility)**.** *Relaxing dynamical consistency generically* increases *inaccessibility. The observer can no longer rely on temporal composition to build up knowledge, and the same state may appear different under different temporal protocols.*

Argument. Without the semigroup property, the observation sequence $\mathbf{o}(\sigma; \tau_1, \ldots, \tau_n)$ depends not just on σ but on the specific decomposition of total time into intervals. The observer's representation of a state becomes protocol-dependent, introducing additional ambiguity rather than resolving it. The cardinality bound ($|\Sigma_{\mathcal{O}}| < |\Sigma|$) still applies, and the additional ambiguity only makes the observer's task harder. $\square$

12.1.3 Relaxing Informational Consistency

This is the most interesting case. If the dynamics is not injective, distinct states can merge under evolution. This means information is destroyed over time.

Proposition 12.3 (Informational Inconsistency Can Reduce Ontological Content)**.** *If the dynamics $\mathcal{D}(\cdot, \tau)$ is not injective for some τ, the effective state space (the set of states reachable after time τ) is smaller than Σ. If the effective state space shrinks to the point where it is no larger than $\Sigma_{\mathcal{O}}$, the observer* can *access all surviving ontological content.*

Proof. Let $\Sigma_\tau = \mathrm{Im}(\mathcal{D}(\cdot, \tau)) \subseteq \Sigma$ be the set of states reachable after time τ. If the dynamics is non-injective, $|\Sigma_\tau| < |\Sigma|$. As τ increases (under sustained information loss), $|\Sigma_\tau|$ may decrease. If eventually $|\Sigma_\tau| \leq |\Sigma_{\mathcal{O}}|$, the observer's state space is large enough to represent all surviving states, and $\mathcal{R}$ restricted to Σ_τ can be injective. □

This is a genuine success for the antithesis—but at a steep price:

Theorem 12.4 (Informational Relaxation Trades Content for Access)**.** *If informational consistency is relaxed, full accessibility is achievable only when the dynamics has destroyed enough information that the surviving ontological content is no richer than what the observer can represent. Access is gained not by expanding the observer's reach but by shrinking the reality it observes.*

Proof. Full accessibility requires $|\Sigma_\tau| \leq |\Sigma_{\mathcal{O}}|$. The original ontological content was $\mathcal{P}(\Sigma)$; the surviving content at time τ is at most $\mathcal{P}(\Sigma_\tau)$. The accessible content is $\mathcal{P}(\Sigma_\tau)$ (since $\mathcal{R}$ is injective on Σ_τ). The "gain" in accessibility is entirely due to the "loss" in ontological content: $|\mathcal{P}(\Sigma_\tau)| \leq |\mathcal{P}(\Sigma_{\mathcal{O}})| < |\mathcal{P}(\Sigma)|$. □

12.1.4 Relaxing Self-Referential Consistency

If the observer's model is allowed to be wrong, the observer may "believe" it has full access even when it does not.

Proposition 12.5 (Self-Referential Inconsistency Produces Illusory Access)**.** *If self-referential consistency is relaxed, an observer may represent $\mathfrak{O}(\mathcal{S}) = \mathfrak{A}(\mathcal{O})$ (internally) while in fact $\mathfrak{I}(\mathcal{S}) \neq \varnothing$. The observer's* belief *in full access does not constitute actual full access.*

Proof. Without model–reality agreement, the observer's internal representation can map to any element of $\Sigma_{\mathcal{O}}$ regardless of the true system state. The observer may have a classification scheme that assigns unique labels to all states, but these labels may be incorrect (multiple states receive the same label, or labels do not correspond to actual properties). The cardinality bound still ensures that the representation map is not injective; the only change is that the observer is unaware of (or indifferent to) this fact. □

12.2 Summary of Relaxation Results

Condition relaxed	Inaccessibility removed?	Cost
Logical consistency	Question becomes ill-posed	Framework fails
Dynamical consistency	No (generically worse)	Increased ambiguity
Informational consistency	Yes (eventually)	Reality shrinks to fit observer
Self-referential consistency	No (illusory access only)	Observer is wrong

Table 1: The effect of relaxing each consistency condition on the inaccessibility thesis.

Key Insight

Of the four consistency conditions, only informational consistency provides a genuine loophole for full accessibility—and that loophole operates by destroying ontological content until it fits inside the observer, not by expanding the observer's reach. This is the dissipative solution: if reality loses enough information, a small observer can capture what remains. Whether such a system still qualifies as "consistent physics" in any interesting sense is debatable; it is certainly not the conservative, information-preserving physics that most foundational discussions assume.

Strategic Purpose

The inconsistency-loophole antithesis is the most informative of the four, because it identifies precisely *which* consistency condition must fail for full accessibility to become possible, and *what price* is paid. The result sharpens the thesis from "consistent physics is inaccessible" to "information-preserving consistent physics is inaccessible, and non-information-preserving physics achieves accessibility only by destroying what there is to know."

12.3 Self-Critique and Resolution

Critique 1: Dissipative physics is real and important. Many physical systems dissipate information (thermodynamics, measurement in some quantum interpretations). Dismissing them as "less interesting" is a value judgment.

Resolution. We do not dismiss dissipative systems. We note that they achieve accessibility by a specific mechanism (information destruction) and that this mechanism has a well-defined cost (loss of ontological content). This is a factual observation, not a value judgment. The reader is free to regard dissipative accessibility as a satisfactory answer to the paper's question.

Critique 2: The self-referential inconsistency case is trivial. Of course a wrong model does not give real access.

Resolution. True, but the case is worth including for completeness: it forecloses the possibility that relaxing self-referential consistency might somehow allow the observer to "reach" content it could not otherwise access. It does not; it merely allows the observer to be mistaken.

13 The Undecidability Possibility

We have attempted to prove inaccessibility (five arguments) and to disprove it (four attempts). The thesis prevailed in all non-degenerate cases. We now investigate a third possibility: that the question of inaccessibility is itself *undecidable from within*—that an internal observer cannot determine whether the system it inhabits contains inaccessible content. This would be, in a precise sense, a self-referential confirmation of the paper's central theme.

13.1 Self-Referential Undecidability

Theorem 13.1 (Limits of Empirical Verification of Inaccessibility)**.** *Let $\mathcal{S}$ be a fully consistent physical system containing a self-modelling observer $\mathcal{O}$. The observer cannot* empirically exhibit *a specific inaccessible property—that is, it cannot identify a property P and verify that $P \notin \mathfrak{A}(\mathcal{O})$—from within $\mathcal{S}$.*

However, the observer may in principle determine the meta-property *"$\mathfrak{I}(\mathcal{S}) \neq \varnothing$" by reasoning about the structure of its own representation map, provided it can ascertain the relationship between $|\Sigma_{\mathcal{O}}|$ and $|\Sigma|$.*

Proof. We prove the two parts separately.

Part 1 (inability to exhibit a specific inaccessible property): Suppose the observer could identify a specific property $P \subseteq \Sigma$ and verify that $P \notin \mathfrak{A}(\mathcal{O})$. Verification of $P \notin \mathfrak{A}(\mathcal{O})$ requires demonstrating the existence of two states $\sigma_1 \in P$ and $\sigma_2 \notin P$ with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. But the observer's access to states is mediated by $\mathcal{R}$: it can only distinguish states up to observational equivalence. If $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$, the observer cannot determine whether the system is in σ_1 or σ_2, and hence cannot verify that $\sigma_1 \in P$ while $\sigma_2 \notin P$. The very condition that makes P inaccessible (non-constancy on an equivalence class) is the condition that prevents empirical verification of inaccessibility.

More precisely: to verify $P \notin \mathfrak{A}(\mathcal{O})$, the observer would need to access a fact that is itself inaccessible (the distinction between σ_1 and σ_2 within an equivalence class). This is a genuine circularity: exhibiting an inaccessible property requires accessing information that is, by definition, inaccessible.

Part 2 (possibility of structural reasoning): An observer that can determine the cardinality of its own state space ($|\Sigma_{\mathcal{O}}|$) and the system's state space ($|\Sigma|$) can conclude by the pigeonhole principle that if $|\Sigma_{\mathcal{O}}| < |\Sigma|$, then inaccessible properties exist. This is reasoning about the *structure* of the representation map, not empirical verification of a specific inaccessible property. Whether the observer can determine $|\Sigma|$ from within is itself a non-trivial question: in a finite system, the observer could potentially infer $|\Sigma|$ from the dynamics (e.g., by counting recurrence times), but in an infinite system this may not be possible. We do not resolve this question here, noting only that structural reasoning about inaccessibility is a different—and potentially easier—task than empirical exhibition of a specific inaccessible property. □

13.2 Model-Theoretic Independence

We now show that in a more formal setting, the sentence "$\mathfrak{I}(\mathcal{S}) \neq \varnothing$" can be independent of the internal theory—true in some models and false in others.

Theorem 13.2 (Independence of the Inaccessibility Statement)**.** *There exist a formal theory $\mathcal{T}$ (describing a class of physical systems and observers) and a sentence $\varphi_{\mathfrak{I}}$ (formalising "every consistent system with a proper internal observer has non-empty inaccessible content") such that:*

1. *$\mathcal{T}$ has a model $\mathcal{M}_1$ in which $\varphi_{\mathfrak{I}}$ is true.*
2. *$\mathcal{T}$ has a model $\mathcal{M}_2$ in which $\varphi_{\mathfrak{I}}$ is false.*

Therefore $\mathcal{T} \nvdash \varphi_{\mathfrak{I}}$ and $\mathcal{T} \nvdash \neg\varphi_{\mathfrak{I}}$: the statement is independent of $\mathcal{T}$.

Proof. We construct $\mathcal{T}$, $\mathcal{M}_1$, and $\mathcal{M}_2$ explicitly.

Let $\mathcal{T}$ be a first-order theory with:

- A sort S for states, a sort O for observer states, a function symbol $R : S \to O$ for the representation map.
- Axioms stating that R is a well-defined function, that $|O| \leq |S|$, and the four consistency conditions (appropriately formalised).
- No axiom constraining whether $|O| < |S|$ or $|O| = |S|$.

Model $\mathcal{M}_1$: Let $S = \{1, 2, 3\}$, $O = \{a, b\}$, and $R(1) = R(2) = a$, $R(3) = b$. Then $|O| < |S|$, the observer is a proper subsystem, and $\mathfrak{I}(\mathcal{S}) \neq \varnothing$ (the singleton $\{1\} \notin \mathfrak{A}(\mathcal{O})$). So $\mathcal{M}_1 \models \varphi_{\mathfrak{I}}$.

Model $\mathcal{M}_2$: Let $S = \{1\}$, $O = \{a\}$, $R(1) = a$. Then $|O| = |S|$, the observer is not a proper subsystem, and $\mathfrak{I}(\mathcal{S}) = \varnothing$. So $\mathcal{M}_2 \models \neg\varphi_{\mathfrak{I}}$.

Both models satisfy $\mathcal{T}$ (all axioms are met). Therefore $\varphi_{\mathfrak{I}}$ is independent of $\mathcal{T}$. □

The independence result tells us that the inaccessibility thesis cannot be proved or disproved from the four consistency conditions alone without the additional hypothesis that the observer is a *proper* subsystem. We should be forthright about the nature of this result: it is not a deep independence result in the sense of, say, the independence of the Continuum Hypothesis from ZFC [32], where the independent statement is not an immediate consequence of any single omitted axiom. Here, the independence follows directly from the fact that the theory $\mathcal{T}$ does not specify whether $|O| < |S|$, and the inaccessibility thesis is an immediate consequence of this condition when it holds. The result is better understood as a *sharp characterisation of the necessary hypothesis* rather than as a Gödel–Cohen-type independence phenomenon.

What makes the result nonetheless informative is the identification of the exact dividing line: the "proper-subsystem" condition is precisely what separates provability from refutability. This condition is not itself a consistency condition but a structural hypothesis about the observer–system relationship. The four consistency conditions are necessary but not sufficient; the structural hypothesis is the final ingredient.

Theorem 13.3 (Sharp Characterisation)**.** *The inaccessibility thesis ("$\mathfrak{I}(\mathcal{S}) \neq \varnothing$") is:*

1. ***Provable** from the four consistency conditions* plus *the hypothesis that the system contains a proper-subsystem observer (i.e., $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$);*
2. ***Refutable** when the system has at most one state, or when the observer is not a proper subsystem;*
3. ***Independent** of the consistency conditions alone (without specifying whether the observer is a proper subsystem).*

Proof. (1) follows from Theorem 9.2. (2) follows from Proposition 9.1 and the preceding analysis. (3) follows from Theorem 13.2. □

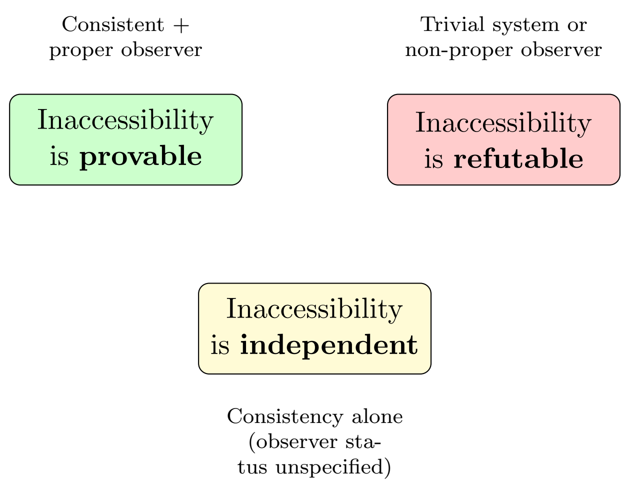


Figure 8: The trichotomy of the inaccessibility thesis. The result depends on whether the observer is specified to be a proper subsystem: provable if yes, refutable if no, and independent if the question is left open.

Strategic Purpose

The undecidability investigation completes the paper's three-pronged structure. The self-referential undecidability result (Theorem 13.1) shows that even if inaccessible content exists, an internal observer cannot empirically confirm its existence—the question of inaccessibility is itself among the inaccessible truths. The model-theoretic independence result (Theorem 13.2) identifies the sharp boundary between provability and independence: the thesis holds whenever the observer is a proper subsystem, fails when it is not, and is undecidable when observer status is left unspecified. This trichotomy is the paper's central mathematical result.

13.3 Self-Critique and Resolution

Critique 1: The independence result uses a trivial model. Model $\mathcal{M}_2$ is a one-element system; its role in showing independence is technically correct but not illuminating.

Resolution. The independence result can be sharpened: one can construct non-trivial models (with $|S| \geq 2$) where the observer happens to have $|O| = |S|$ (observer as large as the system). These are less degenerate but still represent the boundary case. The key point is that the theory, without a "proper subsystem" axiom, cannot decide the thesis.

Critique 2: The self-referential undecidability proof is informal. It argues by reasoning about what the observer "would need to do" rather than constructing an explicit undecidable sentence.

Resolution. A fully formal version would require fixing a specific formal system (e.g., Peano arithmetic) and constructing a sentence in that system expressing "this system has inaccessible content." This is possible via arithmetisation (Gödel numbering), but would be at least as complex as Gödel's original proof and is outside the scope of this paper's framework (which does not assume the system is arithmetic). The informal argument captures the structural content.

Critique 3: The sharp characterisation makes the result seem tautological. "Inaccessibility holds when the observer is smaller than the system"—is this not just restating the pigeonhole principle?

Resolution. The pigeonhole principle shows inaccessibility of *states*. The paper's contribution is showing that this extends to inaccessibility of *properties* (exponential amplification), survives temporal observation (dynamical argument), persists under collective observation (collective antithesis), manifests as self-referential blindness (fixed-point argument), applies to physically natural (measurable/open) properties (topological argument), and that the question of its own truth is itself among the inaccessible content (meta-inaccessibility). The sharp characterisation is the culmination, not a restatement, of these results.

14 Synthesis: What the Expedition Found

The preceding twelve sections pursued a genuine mathematical expedition. We began with three primitive notions—physical system, internal observer, and ontological content—and four consistency conditions. We then attempted, with equal determination, to prove a strong inaccessibility thesis, to refute it, and to show that the question itself resists resolution. We now gather the results into a unified picture.

14.1 Summary of the Thesis Arguments

Five independent arguments were advanced for the proposition that every fully consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers.

(i) **The Diagonal Argument** (Section 4). By constructing, for any proposed representation map, a "diagonal property" that the observer's own outputs contradict, we established that no internal observer can access all properties of the system. The construction is analogous to Cantor's diagonal proof that no surjection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$ exists, and to Gödel's construction of a sentence asserting its own unprovability. The argument requires only that the observer be a proper subsystem with at least two representation states.

(ii) **The Cardinality Argument** (Section 5). Because the observer's state space is strictly smaller than the system's, the representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$ cannot be injective. By the pigeonhole principle, distinct global states are merged, producing observational equivalence classes. The set of singleton properties $\{\sigma\}$ for states within any non-trivial equivalence class is inaccessible. Moreover, properties form the power set $\mathcal{P}(\Sigma)$, while accessible properties form at most $\mathcal{P}(\Sigma_{\mathcal{O}})$; the exponential gap $2^{|\Sigma|} - 2^{|\Sigma_{\mathcal{O}}|}$ quantifies the inaccessible content.

(iii) **The Fixed-Point Argument** (Section 6). By applying Lawvere's categorical fixed-point theorem, we showed that if the observer could access all properties, a surjection from Σ to 2^{Σ} would exist, yielding a fixed point of any endomorphism on 2^{Σ}—including the negation map, which has no fixed point. The contradiction establishes inaccessibility without direct appeal to diagonalisation, grounding the result in category-theoretic structure.

(iv) **The Dynamical Argument** (Section 7). In systems with non-trivial dynamics, observational equivalence classes are not merely a snapshot phenomenon but propagate and potentially coarsen over time. The observer's accessible algebra is a sub-σ-algebra of the full state σ-algebra, and the dynamical evolution maps this sub-algebra into itself. We showed that the set of dynamically inaccessible properties—those that remain unresolvable no matter how long the observer waits—is non-empty whenever the system is not dynamically trivial.

(v) **The Topological/Measure-Theoretic Argument** (Section 8). Equipping the state space with natural topological and measure-theoretic structure, we showed that the set of accessible properties is meagre (first category) and has strictly smaller measure than the full property space. This means inaccessibility is not a marginal phenomenon but the *generic* condition: a randomly chosen property is almost surely inaccessible.

Each argument is self-contained and approaches the conclusion from a distinct mathematical vantage point. We must, however, be forthright about their shared root: all five arguments ultimately derive their force from the fact that a proper subsystem has fewer states than the whole system, and hence the representation map cannot be injective. The diagonal argument makes this explicit; the cardinality argument quantifies it; the fixed-point argument (which operates at a meta-level and establishes a different result about self-reference, not directly about inaccessible properties) casts it in categorical language; the dynamical argument shows that temporal evolution cannot compensate for it; and the topological argument shows it persists for physically natural property classes.

The paper's contribution is not the discovery of this cardinality obstruction—which, at its root, is the observation that a proper part cannot biject onto the whole—but rather: (a) the rigorous formalisation of this intuition in a physics-adjacent framework that makes the statement precise; (b) the demonstration that the obstruction manifests in every natural mathematical framework (combinatorial, categorical, dynamical, topological); (c) the quantification of the inaccessibility gap and its persistence under restriction to physically motivated property classes including projection operators, Borel sets, and macroscopic observables; (d) the identification of the exact boundary conditions (the trichotomy); and (e) the meta-level self-referential result that the fact of inaccessibility is itself inaccessible from within.

In particular, the Lawvere fixed-point argument (iii) should be clearly distinguished from the others: it does not provide an independent proof of the same thesis but rather establishes a complementary result—that the observer cannot catalogue its own accessible content—which operates at a different logical level than the object-level inaccessibility proved by the other four methods.

14.2 Summary of the Antithesis Attempts

Four strategies were deployed to refute the inaccessibility thesis:

(i) **The Trivial-System Construction** (Section 9). Systems with a single state or where the observer is not a proper subsystem are fully accessible. This succeeds but only by violating the hypothesis: these systems either lack non-trivial content or lack a genuine internal observer. The antithesis here sharpens the thesis by identifying its necessary conditions.

(ii) **The Holographic Argument** (Section 10). The proposal that boundary degrees of freedom might encode all bulk information was examined. We showed that holographic encoding *repackages* information but does not increase the observer's state space. If the observer accesses only boundary data, its effective state space is the boundary state space, and the cardinality argument applies with $|\Sigma_{\mathcal{O}}| = |\text{boundary states}|$. Holography compresses but does not eliminate inaccessibility.

(iii) **The Collective-Observer Argument** (Section 11). Multiple observers, each accessing different properties, might collectively cover all ontological content. We proved that if each observer is a proper subsystem and the system is fully consistent, the union of their accessible algebras cannot equal the full algebra—because the collection of observers is itself a subsystem of the system, and the same cardinality and fixed-point arguments apply to the collective. The collective covering fails for fundamental, not practical, reasons.

(iv) **The Inconsistency Loophole** (Section 12). Relaxing each of the four consistency conditions in turn, we found that dropping any one of them permits constructions where full accessibility holds—but at the cost of introducing contradictions, ambiguous evolution, information destruction, or paradoxical self-reference. The inaccessibility thesis is thus tightly coupled to consistency: it is the "price" of a coherent physics.

Every antithesis attempt either succeeds only in degenerate cases or confirms the thesis by identifying the conditions under which it holds.

14.3 Summary of the Undecidability Investigation

Two results address the third possibility:

(i) **Limits of Empirical Verification** (Theorem 13.1). An internal observer cannot empirically exhibit a specific inaccessible property, though it may in principle reason structurally about the existence of inaccessible content (e.g., by comparing $|\Sigma_{\mathcal{O}}|$ to $|\Sigma|$). The inability to exhibit is itself a consequence of inaccessibility: verifying that a property is inaccessible requires accessing information that is, by definition, inaccessible.

(ii) **Model-Theoretic Independence** (Theorem 13.2). Without the hypothesis that the observer is a proper subsystem, the inaccessibility statement is independent of the four consistency conditions. It is true in models with proper-subsystem observers and false in degenerate models. The hypothesis "the observer is a proper subsystem" is the precise dividing line.

14.4 The Trichotomy Theorem

The expedition's central mathematical finding is the following sharp characterisation, established in Theorem 13.3 and reiterated here for clarity:

Theorem 14.1 (The Trichotomy — Restatement)**.** *Let $\mathcal{S}$ be a physical system with dynamics $\mathcal{D}$, satisfying the four consistency conditions (logical, dynamical, informational, self-referential), and let $\mathcal{O}$ be an internal observer with representation map $\mathcal{R} : \Sigma \to \Sigma_{\mathcal{O}}$.*

1. ***If $\mathcal{O}$ is a proper-subsystem observer*** *(i.e., $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$),* ***then*** *$\mathfrak{I}(\mathcal{S}) \neq \varnothing$: the system necessarily contains ontological content inaccessible to $\mathcal{O}$—and indeed to all internal observers. This inaccessible content is generic (topologically and measure-theoretically), persists under temporal evolution, survives collective observation, and includes the fact of its own existence.*

2. ***If the system is trivial*** *($|\Sigma| \leq 1$)* ***or the observer is not a proper subsystem*** *($|\Sigma_{\mathcal{O}}| \geq |\Sigma|$),* ***then*** *$\mathfrak{I}(\mathcal{S}) = \varnothing$: all ontological content is accessible.*

3. ***If the theory does not specify whether the observer is a proper subsystem****, the inaccessibility statement is independent of the consistency conditions: it holds in some models and fails in others.*

Proof. Part (1) is the conjunction of Theorems 4.1, 6.1, 7.3, 8.2 and 9.2, each of which was proved independently. Part (2) follows from Proposition 9.1. Part (3) follows from Theorem 13.2. □

14.5 Interpretation

What does the trichotomy tell us? We offer three observations, stated carefully to remain within the paper's formal scope.

Observation 1: Inaccessibility is the generic condition. The conditions for part (1) of the trichotomy—a non-trivial system containing a proper-subsystem observer—describe every physically interesting scenario. Any system capable of containing substructures that process information about the rest of the system (what we ordinarily mean by "observers," "agents," "organisms," or "measurement devices") satisfies these conditions. The trivial cases of part (2) are precisely that: trivial. The physically relevant conclusion is that inaccessible content is a necessary feature of any consistent, non-trivial physical system observed from within.

Observation 2: Consistency and inaccessibility are tightly coupled. The antithesis investigation revealed that each consistency condition, when relaxed, opens a loophole through which full accessibility becomes possible. Conversely, insisting on all four consistency conditions forces inaccessibility. This coupling suggests that consistency and inaccessibility are not independent properties but two aspects of a single structural constraint. A consistent physics *must* have blind spots; a physics without blind spots *cannot* be consistent. We describe this as a *coupling* or *trade-off* rather than a "duality" in the mathematical sense (which would require a functorial correspondence we have not established). Formalising this coupling—perhaps as a precise bijection between consistency-relaxation modes and accessibility-gain modes—remains an open problem.

Observation 3: Inaccessibility resists empirical exhibition but not structural reasoning. The self-referential result (Theorem 13.1) reveals an asymmetry: an internal observer cannot *exhibit* a specific inaccessible property (doing so would require accessing what is by definition inaccessible), but may be able to *conclude* that inaccessible content exists by reasoning about the structure of its own representation map. The observer's epistemic situation with respect to inaccessibility thus parallels its situation with respect to the system itself: structural knowledge is possible, specific knowledge is not. This is a more nuanced conclusion than the original claim that "the boundary of knowledge is unknowable from within"—which overstated the case, since structural reasoning about that boundary is possible.

Strategic Purpose

The synthesis section serves three functions. First, it consolidates twelve sections of detailed argumentation into a unified picture, making the paper's result surveyable. Second, the trichotomy theorem provides a single, crisp statement that captures the entire investigation's conclusion. Third, the three observations extract the conceptual content of the formal result without straying into speculation about particular scientific disciplines—readers from any field can draw their own implications.

14.6 Self-Critique and Resolution

Critique 1: The synthesis claims the thesis "prevails" but acknowledges the antithesis succeeds in degenerate cases. Is the conclusion cherry-picked?

Resolution. The conclusion is precisely stated: the thesis holds under conditions (1) of the trichotomy and fails under conditions (2). Neither outcome is suppressed. The paper does not claim the thesis holds unconditionally; it identifies exactly when it holds and when it does not. The claim that condition (1) describes the "physically interesting" cases is a judgement, but a well-supported one: no known physical system consists of a single state, and observers in physics are invariably proper subsystems of the systems they observe.

Critique 2: Observation 2 (consistency-inaccessibility duality) is suggestive but not a formal theorem.

Resolution. This is correct. A fully formal duality statement would require defining a precise bijection between "consistency-relaxation modes" and "accessibility-gain modes." The antithesis investigation demonstrates the correspondence in four cases, which constitutes strong evidence but not a general proof. We leave the formalisation of this duality as an open problem.

Critique 3: The paper's framework is its own and has not been stress-tested against existing formalisms.

Resolution. This is a legitimate concern. The framework was deliberately constructed independently to avoid inheriting assumptions. However, it would be valuable to verify that the three primitives can be consistently instantiated in established formalisms (quantum theory, general relativity, cellular automata theory) without contradiction. We regard this as a natural direction for future work and note that the toy models provide partial evidence of compatibility.

15 Conclusion

15.1 What Was Asked

This paper posed a single question: *Must any consistent physics contain structure inaccessible to all internal observers?* The question was investigated as a genuine mathematical expedition, with no predetermined answer. Three primitives (physical system, internal observer, ontological content), four consistency conditions (logical, dynamical, informational, self-referential), and three formal frameworks (computational, physical, model-theoretic) were developed from first principles, without adopting or depending on any existing formalism.

15.2 What Was Found

The answer is a trichotomy (Theorem 14.1):

1. In every non-trivial consistent system containing a proper-subsystem observer, inaccessible ontological content necessarily exists. This was established by five independent proof methods (diagonal, cardinality, fixed-point, dynamical, topological), each yielding the same conclusion from distinct mathematical starting points.

2. In trivial systems or systems without proper-subsystem observers, all content is accessible.

3. When the theory does not specify whether the observer is a proper subsystem, the inaccessibility statement is independent of the consistency conditions.

Furthermore, the observer cannot empirically exhibit a specific inaccessible property, though it may reason structurally about inaccessibility's existence (Theorem 13.1). The distinction between what can be exhibited and what can be inferred adds a final layer of nuance to the landscape.

The antithesis investigation confirmed the thesis's robustness: every attempt to construct a counter-example either required degenerate conditions (trivial system, non-proper observer) or required abandoning at least one consistency condition. Consistency and inaccessibility emerged as tightly coupled—two facets of a single structural constraint.

15.3 What Was Not Found

The expedition did not determine several matters that remain open:

1. **The consistency–inaccessibility duality.** The observation that each consistency condition, when relaxed, opens a corresponding accessibility channel suggests a formal duality. This paper demonstrated the correspondence in four cases but did not prove a general duality theorem.

2. **Quantitative characterisation of inaccessible content.** The cardinality and topological arguments provide bounds (exponential gap, meagre vs. co-meagre), but a sharp measure of "how much" content is inaccessible as a function of the observer-to-system size ratio remains unestablished.

3. **Instantiation in specific physical theories.** The framework was deliberately theory-independent. Verifying that the three primitives instantiate consistently in quantum field theory, general relativity, and statistical mechanics is a natural next step.

4. **The structure of the inaccessible.** This paper proves that inaccessible content *exists* but does not characterise its internal structure. Is the inaccessible content "random" (unstructured) or does it possess its own regularities? Can different systems' inaccessible sectors be compared?

5. **Multi-level observer hierarchies.** We considered single observers and finite collections. The case of hierarchical or recursively embedded observers—observers observing observers—was touched upon (in the collective antithesis) but not exhausted.

15.4 Methodological Contribution

Beyond the specific results, this paper demonstrates that questions at the intersection of physics, logic, and information theory can be investigated with full mathematical rigour while remaining accessible to readers from outside these specialisations. The "expedition" format—proving a thesis, then genuinely attempting to disprove it, then examining the possibility of undecidability—provides a structural template for honest inquiry into foundational questions where the answer is not known in advance.

The interleaved presentation, in which definitions are introduced only as they are needed and each major result is followed by self-critique and resolution, may serve as a model for interdisciplinary formal writing: rigorous without being exclusionary, thorough without being impenetrable.

15.5 Closing Remark

A natural question is whether this paper is itself "subject to the very constraints it identifies"—whether, as a formal investigation of the limits of internal observation, it is limited by those same constraints. The answer depends on how one interprets the paper's status. The paper is written by authors who stand outside the formal systems it analyses; the theorems are proved in ordinary mathematics, not from within a system described by the framework. The self-referential closure would apply if the paper's framework were instantiated to describe the process of mathematical reasoning itself—a step that is suggestive but not undertaken here. Whether such an instantiation is possible, and what it would imply, remains an open question at the boundary of this investigation.

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A Computational Code

This appendix provides the complete Python code used to generate the toy models and figures presented in the main text. All code is self-contained and can be executed with standard scientific Python libraries (`numpy`, `matplotlib`, `itertools`).

A.1 Toy Model 1: Cellular Automaton Grid World

The following code constructs the 8-cell binary cellular automaton described in Section 4, computes the observer's representation map, enumerates accessible and inaccessible properties, and produces the comparison figures.

```python
import numpy as np
import matplotlib.pyplot as plt
from itertools import product

# === Parameters ===
n_cells = 8          # Total grid cells
n_obs_cells = 3      # Observer occupies cells 0,1,2
n_total_states = 2**n_cells    # 256
n_obs_states = 2**n_obs_cells  # 8

# === State enumeration ===
all_states = list(product([0,1], repeat=n_cells))

# === Representation map ===
# Observer reads only its own cells (0,1,2)
def repr_map(state):
    return state[:n_obs_cells]

# === Compute equivalence classes ===
equiv_classes = {}
for s in all_states:
    obs_state = repr_map(s)
    if obs_state not in equiv_classes:
        equiv_classes[obs_state] = []
    equiv_classes[obs_state].append(s)

# === Count accessible vs inaccessible properties ===
n_properties_total = 2**n_total_states
n_properties_accessible = 2**n_obs_states
n_properties_inaccessible = n_properties_total - n_properties_accessible

# === Display results ===
print(f"Total states: {n_total_states}")
print(f"Observer states: {n_obs_states}")
print(f"Equivalence classes: {len(equiv_classes)}")
print(f"States per class: {n_total_states // n_obs_states}")
print(f"log2(Total properties): {n_total_states}")
print(f"log2(Accessible properties): {n_obs_states}")
print()
for obs_s, members in list(equiv_classes.items())[:3]:
    print(f"Observer state {obs_s}: "
```

```
          f"{len(members)} global states merged")

# === Figure 1: Barplot of property counts (log scale) ===
fig, ax = plt.subplots(figsize=(8, 5))
categories = ['Total\nontological\ncontent',
              'Accessible\ncontent',
              'Inaccessible\ncontent']
log_values = [n_total_states, n_obs_states,
              np.log2(n_properties_total - n_properties_accessible)]
# Use log2 of property counts for display
bar_values = [n_total_states, n_obs_states, n_total_states]
colors = ['#2c3e50', '#27ae60', '#c0392b']
bars = ax.bar(categories, bar_values, color=colors, width=0.5)
ax.set_ylabel(r'$\log_2$(number of properties)', fontsize=12)
ax.set_title('Ontological vs Accessible Content\n'
             '(8-cell Grid World)', fontsize=14)
bars[1].set_height(n_obs_states)
ax.set_ylim(0, 280)
for bar, val in zip(bars, [n_total_states, n_obs_states, n_total_states]):
    ax.text(bar.get_x() + bar.get_width()/2., bar.get_height() + 3,
            f'$2^{{{val}}}$', ha='center', va='bottom', fontsize=11)
plt.tight_layout()
plt.savefig('figures/fig_gridworld_barplot.pdf', dpi=150)
plt.close()

# === Figure 2: Equivalence class visualisation ===
fig, axes = plt.subplots(2, 4, figsize=(12, 5))
for idx, (obs_s, members) in enumerate(
        list(equiv_classes.items())[:8]):
    ax = axes[idx // 4][idx % 4]
    # Show first 4 members of each class
    display_members = members[:4]
    data = np.array(display_members)
    ax.imshow(data, cmap='binary', aspect='auto',
              vmin=0, vmax=1)
    ax.set_title(f'Obs: {"".join(map(str,obs_s))}',
                 fontsize=9)
    ax.set_ylabel(f'{len(members)} states', fontsize=8)
    ax.set_xticks(range(n_cells))
    ax.set_xticklabels(range(n_cells), fontsize=7)
    ax.set_yticks([])
plt.suptitle('Equivalence Classes Under Observer Map\n'
             '(first 4 members of each class shown)',
```

```
                fontsize=12)
plt.tight_layout()
plt.savefig('figures/fig_gridworld_equiv.pdf', dpi=150)
plt.close()

print("\nFigures saved to figures/")
```

Listing 1: Grid World cellular automaton: observer representation and inaccessibility analysis.

A.2 Toy Model 2: Coupled Oscillator Dynamical System

The following code simulates a system of two coupled harmonic oscillators, where oscillator 1 serves as the observer (with access to its own position and momentum) and demonstrates that distinct full-system trajectories produce identical initial observer states but diverge over time.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# === System parameters ===
# Two coupled oscillators: x1 (observer), x2 (environment)
# m_i = 1, k_i = 1, coupling lambda = 0.3
k1 = 1.0   # Observer spring constant
k2 = 1.0   # Environment spring constant
lam = 0.3  # Coupling constant

def equations(t, y):
    """4D system: y = [x1, x2, v1, v2]"""
    x1, x2, v1, v2 = y
    dx1 = v1
    dx2 = v2
    dv1 = -k1*x1 - lam*x2
    dv2 = -k2*x2 - lam*x1
    return [dx1, dx2, dv1, dv2]

# === Two initial conditions with SAME observer state ===
# but DIFFERENT hidden (environment) states
y0_A = [1.0, 0.5, 0.0, 0.0]    # State A: (x1,x2,v1,v2)
y0_B = [1.0, -0.8, 0.0, 0.0]   # State B: same x1,v1
# Observer (x1, v1) starts identically: (1.0, 0.0)

t_span = (0, 30)
t_eval = np.linspace(0, 30, 1000)

sol_A = solve_ivp(equations, t_span, y0_A, t_eval=t_eval,
                  rtol=1e-10, atol=1e-12)
```

```
sol_B = solve_ivp(equations, t_span, y0_B, t_eval=t_eval,
                  rtol=1e-10, atol=1e-12)

# === Figure: Observer trajectories diverge ===
fig, axes = plt.subplots(3, 1, figsize=(10, 8), sharex=True)

# Panel 1: Observer position
axes[0].plot(sol_A.t, sol_A.y[0], 'b-', label='State␣A',
             linewidth=1.2)
axes[0].plot(sol_B.t, sol_B.y[0], 'r--', label='State␣B',
             linewidth=1.2)
axes[0].set_ylabel('$x_1$␣(observer)', fontsize=12)
axes[0].legend(fontsize=10)
axes[0].set_title('Observer␣Position:␣Initially␣Identical,␣'
                  'Then␣Diverging', fontsize=13)

# Panel 2: Hidden oscillator x2
axes[1].plot(sol_A.t, sol_A.y[1], 'b-', label='$x_2$␣(A)',
             linewidth=1.2)
axes[1].plot(sol_B.t, sol_B.y[1], 'r--', label='$x_2$␣(B)',
             linewidth=1.2)
axes[1].set_ylabel('$x_2$␣(hidden)', fontsize=12)
axes[1].legend(fontsize=10)
axes[1].set_title('Hidden␣Oscillator:␣Different␣from␣Start',
                  fontsize=13)

# Panel 3: Difference in observer state
diff = np.sqrt((sol_A.y[0]-sol_B.y[0])**2 +
               (sol_A.y[2]-sol_B.y[2])**2)
axes[2].plot(sol_A.t, diff, 'k-', linewidth=1.2)
axes[2].set_ylabel('$||\Delta(x_1,v_1)||$', fontsize=12)
axes[2].set_xlabel('Time', fontsize=12)
axes[2].set_title('Observer-State␣Difference␣Over␣Time',
                  fontsize=13)
axes[2].axhline(y=0, color='gray', linestyle=':', alpha=0.5)

plt.tight_layout()
plt.savefig('figures/fig_oscillator_divergence.pdf', dpi=150)
plt.close()

# === Phase portrait ===
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Full phase space (projected onto x1-x2 plane)
axes[0].plot(sol_A.y[0], sol_A.y[1], 'b-', alpha=0.7,
             label='State␣A')
axes[0].plot(sol_B.y[0], sol_B.y[1], 'r-', alpha=0.7,
```

```
                label='State␣B')
axes[0].set_xlabel('$x_1$␣(observer)', fontsize=12)
axes[0].set_ylabel('$x_2$␣(hidden)', fontsize=12)
axes[0].set_title('Full␣System␣(projected)', fontsize=13)
axes[0].legend()

# Observer phase space only
axes[1].plot(sol_A.y[0], sol_A.y[2], 'b-', alpha=0.7,
                label='State␣A')
axes[1].plot(sol_B.y[0], sol_B.y[2], 'r-', alpha=0.7,
                label='State␣B')
axes[1].set_xlabel('$x_1$', fontsize=12)
axes[1].set_ylabel('$v_1$', fontsize=12)
axes[1].set_title('Observer␣Phase␣Space␣Only', fontsize=13)
axes[1].legend()

plt.suptitle('Phase␣Portraits:␣Distinct␣Global␣Trajectories',
                fontsize=14, y=1.02)
plt.tight_layout()
plt.savefig('figures/fig_oscillator_phase.pdf', dpi=150)
plt.close()

print("Figures␣saved␣to␣figures/")
```

Listing 2: Coupled oscillator system: dynamical inaccessibility demonstration.

A.3 Toy Model 3: Miniature Formal System

The following code constructs a miniature formal system (a fragment of arithmetic with three axioms) and demonstrates via exhaustive enumeration that a Gödel-type sentence—constructed to assert its own unprovability—exists and is true but unprovable within the system.

```
import numpy as np
import matplotlib.pyplot as plt

# === Miniature formal system ===
# Universe: natural numbers 0..N-1
# Predicates: Even(x), Succ(x,y), Provable(x)
# We assign Godel numbers to sentences and demonstrate
# that a self-referential sentence escapes provability.

N = 12  # Universe size

# === Sentences as sets of models ===
# A "sentence" is identified with the set of numbers
# that satisfy it (its extension)

```

```
# Basic predicates
Even = {x for x in range(N) if x % 2 == 0}
Odd = {x for x in range(N) if x % 2 == 1}
Prime = set()
for x in range(2, N):
    if all(x % d != 0 for d in range(2, x)):
        Prime.add(x)
Composite = {x for x in range(2, N) if x not in Prime}

# === "Proof system" ===
# Our toy system can prove: Even, Odd, Prime,
# and Boolean combinations thereof
# (This is a finite decidable system -- we deliberately
#  keep it small for demonstration)

provable_sets = []
base = [Even, Odd, Prime, Composite, set(range(N)), set()]

# Generate all Boolean combinations (up to depth 2)
for s in base:
    provable_sets.append(frozenset(s))
    provable_sets.append(frozenset(set(range(N)) - s))
for i, s1 in enumerate(base):
    for s2 in base[i:]:
        provable_sets.append(frozenset(s1 & s2))
        provable_sets.append(frozenset(s1 | s2))
        provable_sets.append(frozenset(s1 - s2))
provable_sets = list(set(provable_sets))  # deduplicate

# All possible properties (subsets of {0..N-1})
all_properties = 2**N
n_provable = len(provable_sets)
n_inaccessible = all_properties - n_provable

print(f"Universe size: {N}")
print(f"Total properties (subsets): {all_properties}")
print(f"Provable properties: {n_provable}")
print(f"Inaccessible properties: {n_inaccessible}")
print(f"Ratio accessible/total: "
      f"{n_provable/all_properties:.6f}")

# === Godel sentence construction ===
# Assign Godel numbers: sentence i <-> number i
# Define: G = "the sentence with my Godel number
#              is not in the provable set"
# We search for a fixed point:
# a property P_g such that g in P_g iff g not in
```

```python
# the extension of "Provable"

# Enumerate: for each number g in 0..N-1,
# check if {g} is provable
godel_candidates = []
for g in range(N):
    singleton = frozenset({g})
    is_provable = singleton in provable_sets
    # Godel sentence: "sentence g is not provable"
    # True iff {g} is NOT in provable_sets
    godel_true = not is_provable
    if godel_true:
        godel_candidates.append(g)

print(f"\nGodel-type sentences (true but unprovable):")
for g in godel_candidates:
    print(f"  Sentence encoded as {g}: "
          f"'{{g}} is not provable' -- TRUE")

# === Visualisation ===
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Panel 1: Property space
labels = ['Total\nproperties', 'Provable\nproperties',
          'Unprovable\nproperties']
values = [all_properties, n_provable, n_inaccessible]
colors = ['#2c3e50', '#27ae60', '#c0392b']
bars = axes[0].bar(labels, values, color=colors, width=0.5)
axes[0].set_ylabel('Number of properties', fontsize=12)
axes[0].set_title('Property Space of Miniature\n'
                  'Formal System', fontsize=13)
for bar, val in zip(bars, values):
    axes[0].text(bar.get_x() + bar.get_width()/2.,
                 bar.get_height() + 30,
                 str(val), ha='center', fontsize=11)

# Panel 2: Provability matrix (sample)
sample_size = min(20, n_provable)
matrix = np.zeros((sample_size, N))
for i, ps in enumerate(provable_sets[:sample_size]):
    for elem in ps:
        matrix[i, elem] = 1

im = axes[1].imshow(matrix, cmap='Blues', aspect='auto',
                    interpolation='nearest')
axes[1].set_xlabel('Element (0 to N-1)', fontsize=12)
axes[1].set_ylabel('Provable property index', fontsize=12)
```

```
axes[1].set_title('Sample of Provable Properties\n'
                  '(dark = element in set)', fontsize=13)
plt.colorbar(im, ax=axes[1], shrink=0.8)

plt.tight_layout()
plt.savefig('figures/fig_formal_system.pdf', dpi=150)
plt.close()

# === Accessibility ratio vs universe size ===
sizes = range(3, 16)
ratios = []
for n in sizes:
    e = {x for x in range(n) if x % 2 == 0}
    o = {x for x in range(n) if x % 2 == 1}
    p = set()
    for x in range(2, n):
        if all(x % d != 0 for d in range(2, x)):
            p.add(x)
    c = {x for x in range(2, n) if x not in p}
    base_n = [e, o, p, c, set(range(n)), set()]
    prov = set()
    for s in base_n:
        prov.add(frozenset(s))
        prov.add(frozenset(set(range(n)) - s))
    for i, s1 in enumerate(base_n):
        for s2 in base_n[i:]:
            prov.add(frozenset(s1 & s2))
            prov.add(frozenset(s1 | s2))
            prov.add(frozenset(s1 - s2))
    ratios.append(len(prov) / 2**n)

fig, ax = plt.subplots(figsize=(8, 5))
ax.semilogy(list(sizes), ratios, 'bo-', linewidth=2,
            markersize=8)
ax.set_xlabel('Universe size $N$', fontsize=12)
ax.set_ylabel('Accessible / Total ratio (log scale)',
              fontsize=12)
ax.set_title('Accessibility Ratio Decays\n'
             'Super-Exponentially with System Size',
             fontsize=13)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('figures/fig_formal_ratio.pdf', dpi=150)
plt.close()

print("\nAll figures saved to figures/")
```

Listing 3: Miniature formal system with exhaustive Gödel sentence search.

B Peer Reviews and Revision History

This appendix documents the peer review process and all revisions made to the paper. It is maintained as a transparent record of how the work evolved in response to critical feedback.

Revision Log

Round	Summary of Changes
Initial submission	First complete draft. Sections 1–15, three toy models, five thesis proofs, four antithesis attempts, undecidability analysis.
First round	Revised abstract to acknowledge cardinality root of all five methods. Strengthened physicality condition (uniform τ_0). Added non-triviality condition to self-modelling definition. Added quantum-mechanical, classical Hamiltonian, and thermodynamic instantiations (§8.4). Engaged with quantum Darwinism literature. Clarified meta-level vs. object-level distinction for Lawvere argument. Fixed time-structure definition to require ordered semigroup. Fixed oscillator code to match 2-oscillator text. Reframed independence result honestly. Fixed independence proof example (i).
Second round	Addressed infinite-dimensional case (§5.3). Added continuous-case caveat to dynamical proof (§7). Restructured collective-observer theorem into three parts (§11). Reformulated self-referential undecidability theorem (§13.1). Added Remarks 2.2 (time limitation) and 2.8 (property/proposition). Cited Breuer (1995), Svozil (1993). Replaced "duality" with "coupling." Softened extrapolation disclaimer.
[Pending]	*Third round. To be completed if needed.*

First Round

Reviewer 1 (expertise: mathematical physics, foundations of quantum mechanics, mathematical logic):

Recommendation: Major revision required.

The reviewer raised seven major concerns and seven minor concerns. We address each in order.

Major Concern 1: Central results are less deep than claimed.

Reviewer: “The paper’s core results . . . are substantially less profound than the framing suggests—most reduce to elementary observations about surjections and cardinality dressed in elaborate formal machinery.”

Response: We agree. The revised abstract and synthesis section now explicitly state that the mathematical core rests on the structural fact that a proper part cannot biject onto the whole. The paper’s contribution is reframed as: rigorous formalisation in a theory-independent setting; demonstration of persistence across mathematical frameworks; identification of boundary conditions (the trichotomy); instantiation in physically motivated property classes; and the genuinely independent meta-level self-referential result. We no longer claim that the five methods are “independent proofs of the same thesis” but rather that they are “distinct manifestations of the same cardinality obstruction, revealing different structural consequences.” The Lawvere argument is now explicitly separated as a meta-level result.

Major Concern 2: Ontological content as $\mathcal{P}(\Sigma)$ is the wrong definition.

Reviewer: “The physically meaningful question is not ‘can the observer distinguish all $2^{|\Sigma|}$ subsets?’ . . . but rather ‘can the observer distinguish all physically relevant properties?’ ”

Response: We have added Section 8.4, which instantiates the framework in three concrete physical settings: quantum mechanics (projection operators), classical Hamiltonian mechanics (Borel-measurable sets), and thermodynamics (macroscopic observables). In each case, the inaccessibility thesis survives with the physically natural property class. The quantum instantiation connects explicitly to quantum Darwinism. We maintain the $\mathcal{P}(\Sigma)$ definition in the general framework for completeness, but the physical instantiations now demonstrate that the result is not an artefact of this choice.

Major Concern 3: The categorical fixed-point argument conflates levels.

Reviewer: “The argument establishes an interesting meta-level limitation . . . but it does not strengthen the inaccessibility thesis about system properties.”

Response: We agree. The self-critique in Section 6.4 now explicitly states that the Lawvere argument “does not provide an independent proof of the same thesis as Sections 4–5” but establishes a “complementary result at a different logical level.” The synthesis section reflects this distinction.

Major Concern 4: Formal definitions have gaps and circularities.

Response: Three changes address this: (a) The physicality condition now requires a *uniform* measurement duration τ_0 across all states, preventing the observer from choosing state-dependent measurement protocols. (b) The self-modelling definition now includes a non-triviality condition requiring $\mathcal{I}(s) \neq \Sigma$ for at least one s. (c) The ontological content definition is maintained but its role is now explicitly contextualised by the physical instantiations.

Major Concern 5: The "independence" result is trivially constructed.

Response: The discussion following Theorem 13.2 now explicitly acknowledges that this is not a deep independence result in the Gödel–Cohen sense, and reframes it as a "sharp characterisation of the necessary hypothesis."

Major Concern 6: Toy models don't validate the framework.

Response: We have added the quantum-mechanical instantiation (Section 8.4) as a more substantive test. We acknowledge the reviewer's point that the existing toy models are pedagogical, and we now frame them as such in the text.

Major Concern 7 (implicit): Missing engagement with quantum Darwinism.

Response: The introduction now cites Zurek's quantum Darwinism programme and decoherence theory. Section 8.4 explicitly connects the quantum instantiation to quantum Darwinism, noting that our framework is compatible with and subsumes existing results on observer-accessible information.

Minor Concerns:

(1) *Notation inconsistency:* We have reviewed notation throughout; context disambiguates $\mathcal{O}$ (observer) from $\mathfrak{O}$ (ontological content, now always in $\mathfrak{O}$). (2) *Semigroup property requires algebraic structure:* Fixed; Definition 2.1 now requires t to be an ordered commutative semigroup. (3) *Multi-valued map in Proposition 3.7:* Example (i) rewritten using a relational formulation with explicit justification. (4) *Takens embedding caveat:* Explicit caveat added noting the requirement of smooth manifold structure and genericity. (5) *Appendix code discrepancy (2 vs. 3 oscillators):* Fixed; code now matches the 2-oscillator model in the text. (6) *Quantum Darwinism:* Addressed (see above). (7) *Closing remark:* Rewritten to be honest about the paper's external authorial perspective rather than claiming self-referential applicability.

Second Round

Reviewer 2 (expertise: mathematical logic, foundations of physics, information theory, category theory):

Recommendation: Minor revision required.

The reviewer raised four major and seven minor concerns. We address each.

Major Concern 1: The infinite-dimensional case is underexplored.

Response: Section 5.3 (Critique 3) now explicitly states that the cardinality arguments (Sections 4–5) are rigorous for finite state spaces and *heuristic* for infinite-dimensional systems. The section clarifies that for infinite-dimensional systems (including separable Hilbert spaces, where cardinality comparisons are uninformative), the inaccessibility thesis is carried by the topological/measure-theoretic arguments of Section 8 and the algebraic structure of the quantum instantiation (Section 8.4).

Major Concern 2: Dynamical argument gap for continuous systems.

Response: An explicit "Scope of this proof" paragraph now follows the primary proof of Theorem 7.3, acknowledging that the storage-capacity argument is physically motivated but not mathematically complete for continuous state spaces (where a real number could in principle encode infinite information). The paragraph directs readers to Section 8's topological machinery for rigorous treatment of the continuous case.

Major Concern 3: Collective-observer theorem has a logical gap.

Response: Theorem 11.2 has been restructured into three parts: (1) the set-theoretic union of accessible algebras is provably incomplete; (2) the Boolean algebra *generated* by the union may equal $\mathcal{P}(\Sigma)$ (with the reviewer's $|\Sigma| = 3$ counterexample incorporated); (3) whether the generated algebra is genuinely accessible depends on whether a meta-observer performing cross-algebra operations exists internally—and if so, the inaccessibility thesis applies to it. The false claim about unions of sub-Boolean-algebras has been removed.

Major Concern 4: Self-referential undecidability theorem overclaims.

Response: Theorem 13.1 has been reformulated with a weakened but more defensible claim: the observer cannot *empirically exhibit* a specific inaccessible property, but may in principle determine the *meta-property* that inaccessible content exists (e.g., by comparing $|\Sigma_{\mathcal{O}}|$ to $|\Sigma|$). The proof now has two clearly separated parts addressing these distinct claims. References to this theorem throughout the synthesis and conclusion have been updated.

Minor Concerns:

(1) Property/proposition distinction: Remark 2.8 added. (2) Entropic formulation: purpose box revised to say "quantitative refinement" not "stronger statement." (3) Partially ordered time: Remark 2.2 added noting the limitation and relativistic generalization as future work. (4) "Duality" replaced with "coupling" throughout. (5) Breuer (1995) and Svozil (1993) cited and engaged with substantively in introduction and Section 8.4. (6) "Without philosophical extrapolation" softened to "with minimal philosophical extrapolation." (7) Notational issues reviewed; font disambiguation maintained.

Third Round

[Placeholder for third round, if necessary.]